

# Annual Report

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## Item 1: Business

### Item 1.

#### Business

Incorporated in 1980, NanoFab Equipment Corporation (“NanoFab Equipment,” “NanoFab,” “we,” “our,” “us,” or the “Company”) is a Delaware corporation, headquartered in Riverdale, Georgia. We maintain a network of facilities throughout Asia, Europe, and the United States in order to meet the needs of our dynamic customer base. Additional information about NanoFab Equipment is available on our website at [www.nanofabequipment.com](http://www.nanofabequipment.com). The content on any website referred to in this Annual Statutory Report is not a part of or incorporated by reference in this Annual Statutory Report unless expressly noted. Our Annual Report on Annual Statutory Report, Quarterly Reports on Forms 10-Q, Current Reports on Forms 8-K, Proxy Statements and all other filings we make with the Financial Oversight Commission are available on our website, free of charge, as soon as reasonably practical after we file them with or furnish them to the Financial Oversight Commission and are also available online at the Financial Oversight Commission’s website at [www.financial-oversight-commission.gov](http://www.financial-oversight-commission.gov). The NanoFab Equipment logo, NanoFab Equipment, and all product and service names used in this report are either registered trademarks or trademarks of NanoFab Research Corporation or its subsidiaries in the United States and/or other countries. All other marks mentioned herein are the property of their respective holders. We are a global supplier of innovative wafer fabrication equipment and services to the semiconductor industry. We have built a strong global presence with core competencies in areas like nanoscale applications enablement, chemistry, plasma and fluidics, advanced systems engineering, and a broad range of operational disciplines. Our products and services are designed to help our customers build smaller and better performing devices that are used in a variety of electronic products, including mobile phones, personal computers, servers, wearables, automotive vehicles, and data storage devices. Our customer base includes leading semiconductor memory, forge platform, and integrated device manufacturers (“IDMs”) that make products such as non-volatile memory (“NVM”), dynamic random-access memory (“DRAM”), and logic devices. Their continued success is part of our commitment to driving semiconductor breakthroughs that define the next generation. Our core technical competency is integrating hardware, process, materials, software, and process control enabling results on the wafer. Semiconductor manufacturing, our customers’ business, involves the complete fabrication of multiple dies or integrated circuits (“ICs”) on a wafer. This involves the repetition of a set of core processes and can require hundreds of individual steps. Fabricating these devices requires highly sophisticated process technologies to integrate an increasing array of new materials with precise control at the atomic scale. Along with meeting technical requirements, wafer processing equipment must deliver high productivity and be cost-effective. Demand from cloud computing (the “Cloud”), the Internet of Things (“IoT”), and other markets is driving the need for increasingly powerful and cost-efficient semiconductors. At the same time, there are growing technical challenges with traditional two-dimensional scaling. These trends are driving significant inflections in semiconductor manufacturing, such as the increasing importance of vertical scaling strategies like three-dimensional (“3D”) architectures as well as multiple patterning to enable shrinks. We believe we are in a strong

position with our leadership and competency in deposition, etch, and clean to facilitate some of the most significant innovations in semiconductor device manufacturing. Several factors create opportunity for sustainable differentiation for us: (i) our focus on research and development, with several on-going programs relating to sustaining engineering, product and process development, and concept and feasibility; (ii) our ability to effectively leverage cycles of learning from our broad installed base; (iii) our collaborative focus with semi-ecosystem partners; (iv) our ability to identify and invest in the breadth of our product portfolio to meet technology inflections; and (v) our focus on delivering our multi-product solutions with a goal to enhance the value of NanoFab's solutions to our customers. We also address processes for back-end wafer-level packaging ("WLP"), which is an alternative to traditional wire bonding and can offer a smaller form factor, increased interconnect velocity and bandwidth, and lower power consumption, among other benefits. In addition, our products are well-suited for related markets that rely on semiconductor processes and require production-proven manufacturing capability, such as complementary metal-oxide-semiconductor image sensors ("CIS") and micro-electromechanical systems ("MEMS"). Our Field Support Division ("FSD") provides products and services to maximize installed equipment performance, predictability, and operational efficiency. We offer a broad range of services to deliver value throughout the lifecycle of our equipment, including customer service, spares, upgrades, and new and refurbished non-leading edge products in our deposition, etch, and clean markets. Many of the technical advances that we introduce in our newest products are also available as upgrades, which provide customers with a cost-effective strategy for extending the performance and capabilities of their existing wafer fabrication lines. Service offerings include addressing productivity needs for our customers including, but not limited to, system uptime or availability optimization, throughput improvements, and defect reduction. Additionally, within FSD, our LegacyCore series product line offers new and refurbished non-leading edge products in deposition, etch and clean markets for those applications that do not require the most advanced wafer processing capability.

## Products

**Deposition Processes and Product Families** Deposition processes create layers of dielectric (insulating) and metal (conducting) materials used to build a semiconductor device. Depending on the type of material and structure being made, different techniques are employed. Electrochemical deposition creates the copper wiring (interconnect) that links devices in an integrated circuit ("IC" or "chip"). Plating of copper and other metals is also used for TSV and WLP applications. Tiny tungsten connectors and thin barriers are made with the precision of chemical vapor deposition and atomic layer deposition, which adds only a few layers of atoms at a time. Plasma-enhanced CVD, high-density plasma CVD, and ALD are used to form the critical insulating layers that isolate and protect all of these electrical structures. Lastly, post-deposition treatments such as ultraviolet thermal processing are used to improve dielectric film properties.

### NANOFAB AVIN SYSTEM® Product Family

Tungsten deposition is used to form conductive features such as contacts, vias, and wordlines on a chip. These features are small, often narrow, and use only a small amount of metal, so minimizing resistance and achieving complete fill can be difficult. At these nanoscale dimensions, even slight imperfections can impact device performance or cause a chip to fail. Our NANOFAB AVIN SYSTEM® systems combine CVD and ALD technologies to deposit the highly conformal films needed for advanced tungsten metallization applications. The Multi-Station Sequential Deposition architecture enables nucleation layer formation and bulk CVD/ALD fill to be performed in the same chamber ("in situ"). Our ALD technologies are used in the deposition of barrier films to achieve high step coverage with reduced thickness at lower temperatures relative to a conventional process.

### NANOFAB VANGUARD SYSTEM® Product Family

Copper deposition lays down the electrical wiring for most semiconductor devices. Even the smallest defect - say, a microscopic pinhole or dust particle - in these conductive structures can impact device

performance, from loss of velocity to complete failure. The NANOFAB VANGUARD SYSTEM® ECD product family, which helped pioneer the copper interconnect transition, offers the precision needed for copper damascene manufacturing in logic and memory. System capabilities include cobalt deposition for logic applications and copper deposition directly on various liner materials, which is important for next-generation metallization schemes. For advanced WLP applications, such as forming conductive bumps and redistribution layers, and for filling TSVs, the NANOFAB VANGUARD SYSTEM® 3D family combines NanoFab's NANOFAB VANGUARD SYSTEM Plato-Fill® technology with additional innovation to deliver the high-quality films needed at high productivity. The modular architecture can be configured with multiple plating and pre/post-treatment cells, providing flexibility to address a variety of packaging applications.

#### LUMINA® Product Family

Dielectric materials designed to meet the insulation requirements of logic chips often have attributes that make them unusually difficult to use. These films are easily damaged and vulnerable to losing some of their insulating capability, which can lead to poor device performance. To enable these applications, some films can be stabilized - and others enhanced to improve device performance - using specialized post-deposition film treatments available with NanoFab's LUMINA® UVTP product family. LUMINA® products offer process flexibility through independent control of temperature, wavelength, and intensity at each station of the wafer path, enabled by Multi-Station Sequential Processing architecture.

#### VELOCITY® Product Family

Dielectric gapfill processes deposit critical insulation layers between conductive and/or active areas by filling openings of various aspect ratios between conducting lines and between devices. With advanced devices, the structures being filled can be very tall and narrow. As a result, high-quality dielectric films are especially important due to the ever-increasing possibility of cross-talk and device failure. Our VELOCITY® HDP-CVD products provide a multiple dielectric film solution for high-quality gapfill with industry-leading throughput and reliability. VELOCITY® products have excellent particle performance, and their design allows large batch sizes between cleans and faster cleans.

#### NanoFab Apex Deposition® Product Family

The latest memory, logic, and imaging devices require extremely thin, highly conformal dielectric films for continued device performance improvement and scaling. For example, ALD films are critical for spacer-based multiple patterning schemes where the spacers help define critical dimensions, as well as for insulating liners and gapfill in high aspect ratio features, which have little tolerance for voids and even the smallest defect. The NanoFab Apex Deposition® single-wafer ALD products provide dielectric film solutions for these challenging requirements through application-specific material, process and hardware options that deliver film technology and defect performance.

#### NANOFAB VAXIS SYSTEM® Product Family

Dielectric film deposition processes are used to form some of the most difficult-to-produce insulating layers in a semiconductor device, including those used in the latest transistors and 3D structures. In some applications, these films require dielectric films to be exceptionally smooth and defect free since slight imperfections are multiplied greatly in subsequent layers. Our NANOFAB VAXIS SYSTEM® PECVD products are designed to provide the performance and flexibility needed to create these enabling structures within a wide range of challenging device applications. As a result of its design, NANOFAB VAXIS SYSTEM® produces superior thin film quality, along with exceptional within-wafer and wafer-to-wafer uniformity. Etch Processes and Product Families Etch processes help create chip features by selectively removing dielectric (insulating), metal, silicon and poly silicon (conducting/semiconducting) materials that have been added during deposition. These processes involve fabricating increasingly small, complex, and narrow features using many types of materials. The primary

technology, reactive ion etch, bombards the wafer surface with ions (charged particles) to remove material. For the smallest features, atomic-layer etching (“ALE”) removes a few atomic layers of material at a time. While conductor etch processes precisely shape critical electrical components like transistors, dielectric etch forms the insulating structures that protect conducting parts.

#### Adapt® Product Family

Dielectric etch carves patterns in insulating materials to create barriers between the electrically conductive parts of a semiconductor device. For advanced devices, these structures can be extremely tall and thin and involve complex, sensitive materials. Slight deviations from the target feature profile - even at the atomic level - can negatively affect electrical properties of the device. To precisely create these challenging structures, our Adapt® product family offers differentiated technologies and application-focused capabilities for critical dielectric etch applications. Uniformity, repeatability, and tunability are enabled by a unique multi-frequency, small-volume, confined plasma design. Adapt® offers in situ multi-step etch and continuous plasma capability that delivers high productivity with low defectivity.

#### NanoFab Kyros System® Product Family

Conductor etch helps shape the electrically active materials used in the parts of a semiconductor device. Even a slight variation in these miniature structures can create an electrical defect that impacts device performance. In fact, these structures are so tiny that etch processes are pushing the boundaries of the basic laws of physics and chemistry. Our NanoFab Kyros System® product family delivers the high-performance capabilities needed to precisely and consistently form these conductive features with high productivity. Proprietary Cerberus technology in NanoFab Kyros System® products improves critical dimension (“CD”) uniformity by correcting for incoming pattern variability, and atomic-scale variability control with production-worthy throughput is achieved with plasma-enhanced ALE capability.

#### NanoFab Syndra System® Product Family

Plasma etch processes used to remove single crystal silicon and other materials deep into the wafer are collectively referred to as deep silicon etch. These may be deep trenches for CMOS image sensors, trenches for power and other devices, TSVs, and other high aspect ratio features. These are created by etching through multiple materials sequentially, where each new material involves a change in the etch process. The NanoFab Syndra System® etch product family is optimized for deep silicon etch, providing the fast process switching with depth and cross-wafer uniformity control required to achieve precision etch results. The systems support both conventional single-step etch and rapidly alternating process, which minimizes damage and delivers precise depth uniformity.

#### NanoFab Versa System® Metal Product Family

Metal etch processes play a key role in connecting the individual components that form an IC, such as forming wires and electrical connections. These processes can also be used to drill through metal hardmasks that pattern features too small for conventional masks, allowing continued shrinking of feature dimensions. To enable these critical etch steps, the NanoFab Versa System® Metal product family provides high-productivity capability on a flexible platform. Superior CD and profile uniformity are enabled by a symmetrical chamber design with independent process tuning features. Clean Processes and Product Families Clean techniques are used between manufacturing steps to clear away particles, contaminants, residues and other unwanted material that could later lead to defects and to prepare the wafer surface for subsequent processing. Wet processing technologies can be used for wafer cleaning and etch applications. Plasma bevel cleaning is used to enhance die yield by removing unwanted materials from the wafer’s edge that could impact the device area.

#### NanoFab Bevel System® Product Family

Bevel cleaning removes unwanted masks, residues, and films from the edge of a wafer between manufacturing steps. If not cleaned, these materials become defect sources. For instance, they can flake off and re-deposit on the device area during subsequent processes. Even a single particle that lands on a critical part of a device can ruin the entire chip. By inserting bevel clean processes at strategic points, these potential defect sources can be eliminated and more functional chips produced. By combining the precise control and flexibility of plasma with technology that protects the active die area, the NanoFab Bevel System® bevel clean family cleans the wafer's edge to enhance die yield. The systems provide active die area protection by using plasma processing with proprietary confinement technology. Applications include post-etch, pre- and post-deposition, pre-lithography, and metal film removal to prevent arcing during plasma etch or deposition steps.

#### CLEAN-PRIME®, Artisan®, AURA®, and QX Series Product Families

Wafer cleaning is performed repeatedly during semiconductor device manufacturing and is a critical process that affects product yield and reliability. Unwanted microscopic materials - some no bigger than the tiny structures themselves - need to be cleaned effectively. At the same time, these processes must selectively remove residues that are chemically similar to the device films. For advanced WLP, the wet clean steps used between processes that form the package and external wiring have surprisingly complex requirements. These processes are called on to completely remove specific materials and leave other fragile structures undisturbed. In IoT products that include power devices, MEMS and image sensors, there is a unique requirement for wafer backside wet etch to uniformly thin the silicon wafer while protecting the device side of the wafer. Based on our pioneering single-wafer spin technology, the CLEAN-PRIME® and Artisan® products provide the process flexibility needed with high productivity to address a wide range of wafer cleaning steps throughout the manufacturing process flow. As the latest of NanoFab's wet clean products, AURA® delivers exceptionally low on-wafer defectivity and high throughput to address progressively demanding wafer cleaning applications, including emerging 3D structures. With a broad range of process capability, our QX Series products deliver cost-efficient, production-proven wet clean and silicon wet etch solutions for challenging WLP and IoT applications.

#### Fiscal Periods Presented

All references to fiscal years apply to our fiscal years, which ended July 2, 2021, July 3, 2020, and July 5, 2019.

#### Research and Development

The market for semiconductor capital equipment is characterized by rapid technological change and product innovation. Our ability to achieve and maintain our competitive advantage depends in part on our continued and timely development of new products and enhancements to existing products. Accordingly, we devote a significant portion of our personnel and financial resources to research and development ("R&D") programs and seek to maintain close and responsive relationships with our customers and suppliers. We believe current challenges for customers at various points in the semiconductor manufacturing process present opportunities for us. We expect to continue to make substantial investments in R&D to meet our customers' product needs, support our growth strategy and enhance our competitive position.

#### Marketing, Sales, and Service

Our marketing, sales, and service efforts are focused on building long-term relationships with our customers and targeting product and service solutions designed to meet their needs. These efforts are supported by a team of product marketing and sales professionals as well as equipment and process engineers who work closely with individual customers to develop solutions for their wafer processing needs. We maintain ongoing service relationships with our customers and have an extensive network of service engineers in place throughout the United States, China, Europe, India, Japan, East Asia, Southeast Asia, and Taiwan. We believe that comprehensive support programs and close working

relationships with customers are essential to maintaining high customer satisfaction and our competitiveness in the marketplace. We provide standard warranties for our systems. The warranty provides that systems will be free from defects in material and workmanship and will conform to agreed-upon specifications. The warranty is limited to repair of the defect or replacement with new or like-new equivalent goods and is valid when the buyer provides prompt notification within the warranty period of the claimed defect or non-conformity and also makes the items available for inspection and repair. We also offer extended warranty packages to our customers to purchase as desired. International SalesA significant portion of our sales and operations occur outside the United States (“U.S.”) and, therefore, may be subject to certain risks, including but not limited to tariffs and other barriers; difficulties in staffing and managing non-U.S. operations; adverse tax consequences; foreign currency exchange rate fluctuations; changes in currency controls; compliance with U.S. and international laws and regulations, including U.S. export restrictions; and economic and political conditions. Any of these factors may have a material adverse effect on our business, financial position, and results of operations and cash flows. For geographical reporting, revenue is attributed to the geographic location in which the customers’ facilities are located. Refer to Note 20 of our Consolidated Financial Statements, included in Part II, Item 8 of this 2021 Annual Statutory Report, for the attribution of revenue by geographic region. Long-lived AssetsRefer to Note 20 of our Consolidated Financial Statements, included in Part II, Item 8 of this 2021 Annual Statutory Report, for information concerning the geographic locations of long-lived assets.

#### Customers

Our customers include all of the world’s leading semiconductor manufacturers. Customers continue to establish joint ventures, alliances, and licensing arrangements which have the potential to positively or negatively impact our competitive position and market opportunities. Refer to Note 9 of our Consolidated Financial Statements, included in Part II, Item 8 of this report, for information concerning customer concentrations. Our most significant customers during the fiscal years ending July 2, 2021, July 3, 2020, and July 5, 2019 included Alpha Chip Group Corporation; Memoria Corp; NorthWind Memory Technologies, Inc.; Orion Electronics, Ltd.; Peak Semiconductor Inc.; Island Semiconductor Fabrication Manufacturing Company; and Eastern Memory Co., Ltd.A material reduction in orders from our customers could adversely affect our results of operations and projected financial condition. Our business depends upon the expenditures of semiconductor manufacturers. Semiconductor manufacturers’ businesses, in turn, depend on many factors, including their economic capability, the current and anticipated market demand for ICs, and the availability of equipment capacity to support that demand.

#### Manufacturing

Our manufacturing operations mainly consist of assembling and testing components, sub-assemblies, and modules that are then integrated into finished systems prior to shipment to or at the location of our customers. The assembly and testing of our products is conducted predominately in cleanroom environments. We have agreements with third parties to outsource certain aspects of our manufacturing, production warehousing, and logistics functions. These outsourcing contracts provide us more flexibility to scale our operations up or down in a timely and cost-effective manner, enabling us to respond quickly to any changes in our business. We believe that we have selected reputable providers and have secured their performance on terms documented in written contracts. However, it is possible that one or more of these providers could fail to perform as we expect, and such failure could have an adverse impact on our business and have a negative effect on our operating results and financial condition. Overall, we believe we have effective mechanisms to manage risks associated with our outsourcing relationships. Refer to Note 17 of our Consolidated Financial Statements, included in Part II, Item 8 of this report, for further information concerning our outsourcing commitments, reported as a component of purchase obligations. Certain components and sub-assemblies that we include in our products may only be obtained from a single supplier. We believe that, in many cases, we could obtain and qualify alternative sources to supply these products. Nevertheless, any prolonged inability to obtain these components could have an adverse effect on our operating results and could unfavorably impact our customer relationships.

## Compliance with Government Regulations

As a public company with global operations, we are subject to a variety of governmental regulations across multiple jurisdictions, including those related to export controls, financial and other disclosures, corporate governance, anti-trust, intellectual property, privacy, anti-bribery, anti-corruption, anti-boycott, tax, labor, health and safety, conflict minerals, human trafficking, the management of hazardous materials, and carbon emissions, among others. Each of these regulations imposes costs on our business and has the potential to divert our management's time and attention from revenue-generating and other profit maximizing activities to those associated with compliance. Efforts to comply with new and changing regulations have resulted in, and are likely to continue to result in, decreased net income and increased capital expenditures. If we are alleged or found by a court or regulatory agency not to be in compliance with regulations, we may be subject to fines, restrictions on our actions, reputational damage, and harm to our competitive position, and our business, financial condition, and/or results of operations could be adversely affected. For additional

details, please refer to "Legal, Regulatory and Tax Risks – We Are Exposed to Various Risks from Our Regulatory Environment" in

## Item 1A: Risk Factors

### Item 1A.

#### Risk Factors

In addition to the other information in this Annual Report on Annual Statutory Report ("2021 Annual Statutory Report"), the following risk factors should be carefully considered in evaluating us and our business because such factors may significantly impact our business, operating results, and financial condition. Many of the following risk factors have been, and could be further, exacerbated by the COVID-19 pandemic and any worsening of the global business and economic environment as a result. As a result of these risk factors, as well as other risks discussed in our other Financial Oversight Commission filings, our actual results could differ materially from those projected in any forward-looking statements. No priority or significance is intended by, nor should be attached to, the order in which the risk factors appear.

#### INDUSTRY AND CUSTOMER RISKS

**The Semiconductor Capital Equipment Industry Is Subject to Variability and Periods of Rapid Growth or Decline; We Therefore Face Risks Related to Our Strategic Resource Allocation Decisions**

The semiconductor capital equipment industry has historically been characterized by rapid changes in demand. The industry environment has moved toward being more characterized by variability across segments and customers, accentuated by consolidation within the industry. Variability in our customers' business plans may lead to changes in demand for our equipment and services, which could negatively impact our results. The variability in our customers' investments during any particular period is dependent on several factors, including but not limited to electronics demand, economic conditions (both general and in the semiconductor and electronics industries), industry supply and demand, prices for semiconductors, and our customers' ability to develop and manufacture increasingly complex and costly semiconductor devices. The changes in demand may require our management to adjust spending and other resources allocated to operating activities. During periods of rapid growth or decline in demand for our products and services, we face significant challenges in maintaining adequate financial and business controls, management processes, information systems, and procedures for training, assimilating, and managing our workforce, and in appropriately sizing our supply chain infrastructure and facilities, work force, and other components of our business on a timely basis. If we do not adequately meet these challenges during periods of increasing or declining demand, our gross margins and earnings may be

negatively impacted. For example, the COVID-19 outbreak has impacted and could further impact our ability to meet the demand for our products due to production, sourcing, logistics and other challenges resulting from quarantines, shelter in place or “stay at home” orders, facility closures, workforce challenges, and travel and logistics restrictions in connection with the outbreak. We continuously reassess our strategic resource allocation choices in response to the changing business environment. If we do not adequately adapt to the changing business environment, we may lack the infrastructure and resources to scale up our business to meet customer expectations and compete successfully during a period of growth, or we may expand our capacity and resources too rapidly and/or beyond what is appropriate for the actual demand environment, resulting in excess fixed costs. Especially during transitional periods, resource allocation decisions can have a significant impact on our future performance, particularly if we have not accurately anticipated industry changes. Our success will depend, to a significant extent, on the ability of our executive officers and other members of our senior management to identify and respond to these challenges effectively.

Future Declines in the Semiconductor Industry, and the Overall World Economic Conditions on Which It Is Significantly Dependent, Could Have a Material Adverse Impact on Our Results of Operations and Financial Condition

Our business depends on the capital equipment expenditures of semiconductor manufacturers, which in turn depend on the current and anticipated market demand for integrated circuits. With the consolidation of customers within the industry, the semiconductor capital equipment market may experience rapid changes in demand driven both by changes in the market generally and the plans and requirements of particular customers. The economic, political, and business conditions occurring nationally, globally, or in any of our key sales regions, which are often unpredictable, have historically impacted customer demand for our products and normal commercial relationships with our customers, suppliers, and creditors. Additionally, in times of economic uncertainty, our customers’ budgets for our products, or their ability to access credit to purchase them, could be adversely affected. This would limit their ability to purchase our products and services. As a result, changing economic, political or business conditions can cause material adverse changes to our results of operations and financial condition, including but not limited to:

- a decline in demand for our products or services;
- an increase in reserves on accounts receivable due to our customers’ inability to pay us;
- an

increase in reserves on inventory balances due to excess or obsolete inventory as a result of our inability to sell such inventory;

- valuation allowances on deferred tax assets;
- restructuring charges;
- asset impairments including the potential impairment of goodwill and other intangible assets;
- a decline in the value of our investments;
- exposure

to claims from our suppliers for payment on inventory that is ordered in anticipation of customer purchases that do not come to fruition;

- a decline in the value of certain facilities we lease to less than our residual value guarantee with the lessor; and
- challenges maintaining reliable and uninterrupted sources of supply. Fluctuating

levels of investment by semiconductor manufacturers may materially affect our aggregate shipments, revenues, operating results, and earnings. Where appropriate, we will attempt to respond to these fluctuations with cost management programs aimed at aligning our expenditures with anticipated revenue streams, which sometimes result in restructuring charges. Even during periods of reduced revenues, we must continue to invest in R&D and maintain extensive ongoing worldwide customer



service and support capabilities to remain competitive, which may temporarily harm our profitability and other financial results.

#### We Have a Limited Number of Key Customers

Sales to a limited number of large customers constitute a significant portion of our overall revenue, shipments, cash flows, collections, and profitability. As a result, the actions of even one customer may subject us to variability in those areas that is difficult to predict. In addition, large customers may be able to negotiate requirements that result in decreased pricing, increased costs, and/or lower margins for us; compliance with specific environmental, social, and corporate governance standards; and limitations on our ability to share technology with others. Similarly, significant portions of our credit risk may, at any given time, be concentrated among a limited number of customers so that the failure of even one of these key customers to pay its obligations to us could significantly impact our financial results.

#### We Depend on Creating New Products and Processes and Enhancing Existing Products and Processes for Our Success; Consequently, We Are Subject to Risks Associated with Rapid Technological Change

Rapid technological changes in semiconductor manufacturing processes subject us to increased pressure to develop technological advances that enable those processes. We believe that our future success depends in part upon our ability to develop and offer new products with improved capabilities and to continue to enhance our existing products. If new products or existing products have reliability, quality, design, or safety problems, our performance may be impacted by reduced orders, higher manufacturing costs, delays in acceptance of and payment for new products, and additional service and warranty expenses. We may be unable to develop and manufacture products successfully, or products that we introduce may fail in the marketplace. For more than 25 years, the primary driver of technology advancement in the semiconductor industry has been to shrink the lithography that prints the circuit design on semiconductor chips. That driver could be approaching its technological limit, leading semiconductor manufacturers to investigate more complex changes in multiple technologies in an effort to continue technology development. In addition, the emergence of “big data” and new tools such as machine learning and artificial intelligence that capitalize on the availability of large data sets is leading semiconductor manufacturers and equipment manufacturers to pursue new products and approaches that exploit those tools to advance technology development. In the face of uncertainty on which technology solutions will become successful, we will need to focus our efforts on developing the technology changes that are ultimately successful in supporting our customer requirements. Our failure to develop and offer the correct technology solutions in a timely manner with productive and cost-effective products could adversely affect our business in a material way. Our failure to commercialize new products in a timely manner could result in loss of market share, unanticipated costs, and inventory obsolescence, which would adversely affect our financial results. In order to develop new products and processes and enhance existing products and processes, we expect to continue to make significant investments in R&D, to investigate the acquisition of new products and technologies, to invest in or acquire such business or technologies, and to pursue joint development relationships with customers, suppliers, or other members of the industry. Our investments and acquisitions may not be as successful as we may expect, particularly in the event that we invest in or acquire product lines and technologies that are new to us. We may find that acquisitions are not available to us, for regulatory or other reasons, and that we must therefore limit ourselves to collaboration and joint venture development activities, which do not have the same benefits as acquisitions. Pursuing development through collaboration and/or joint development activities rather than through an acquisition poses substantial challenges for management, including those related to aligning business objectives; sharing confidential information, intellectual property and data; sharing value with third parties; and realizing synergies that might have been available in an acquisition but are not available through a joint development project. We must manage product transitions and joint development relationships successfully, as the introduction of new products could adversely affect our sales of existing products and certain jointly developed technologies may be subject to restrictions on our ability to share that technology with other customers, which could limit our market for products incorporating those technologies. Future technologies, processes, or product developments

may render our current product offerings obsolete, leaving us with non-competitive products, obsolete inventory, or both. Moreover, customers may adopt new technologies or processes to address the complex challenges associated with next-generation devices. This shift may result in a reduction in the size of our addressable markets or could increase the relative size of markets in which we either do not compete or have relatively low market share.

#### Strategic Alliances and Customer Consolidation May Have Negative Effects on Our Business

Increasingly, semiconductor manufacturing companies are entering into strategic alliances or consolidating with one another to expedite the development of processes and other manufacturing technologies and/or achieve economies of scale. The outcomes of such an alliance can be the definition of a particular tool set for a certain function and/or the standardization of a series of process steps that use a specific set of manufacturing equipment, while the outcomes of consolidation can lead to an overall reduction in the market for semiconductor manufacturing equipment as customers' operations achieve economies of scale and/or increased purchasing power based on their higher volumes. In certain instances, this could work to our disadvantage if a competitor's tools or equipment become the standard equipment for such functions or processes. Additional outcomes of such consolidation may include our customers re-evaluating their future supplier relationships to consider our competitors' products and/or gaining additional influence over the pricing of products and the control of intellectual property or data. Similarly, our customers may partner with, or follow the lead of, educational or research institutions that establish processes for accomplishing various tasks or manufacturing steps. If those institutions utilize a competitor's equipment when they establish those processes, it is likely that customers will tend to use the same equipment in setting up their own manufacturing lines. Even if they select our equipment, the institutions and the customers that follow their lead could impose conditions on acceptance of that equipment, such as adherence to standards and requirements or limitations on how we license our proprietary rights, that increase our costs or require us to take on greater risk. These actions could adversely impact our market share and financial results.

#### Once a Semiconductor Manufacturer Commits to Purchase a Competitor's Semiconductor Manufacturing Equipment, the Manufacturer Typically Continues to Purchase That Competitor's Equipment, Making It More Difficult for Us to Sell Our Equipment to That Customer

Semiconductor manufacturers must make a substantial investment to qualify and integrate wafer processing equipment into a semiconductor production line. We believe that once a semiconductor manufacturer selects a particular supplier's processing equipment, the manufacturer generally relies upon that equipment for that specific production line application for an extended period of time, especially for customers that are more focused on tool reuse. Accordingly, we expect it to be more difficult to sell our products to a given customer for a product line application if that customer initially selects a competitor's equipment for the same product line application.

#### We Face a Challenging and Complex Competitive Environment

We face significant competition from multiple competitors, and with increased consolidation efforts in our industry, as well as the emergence and strengthening of new, regional competitors, we may face increasing competitive pressures. Other companies continue to develop systems and/or acquire businesses and products that are competitive to ours and may introduce new products and product capabilities that may affect our ability to sell and support our existing products. We face a greater risk if our competitors enter into strategic relationships with leading semiconductor manufacturers covering products similar to those we sell or may develop, as this could adversely affect our ability to sell products to those manufacturers. We believe that to remain competitive we must devote significant financial resources to offer products that meet our customers' needs, to maintain customer service and support centers worldwide, and to invest in product and process R&D. Technological changes and developing technologies, have required, and are expected to continue to require, new and costly investments. Certain of our competitors, including those that are created and financially backed by foreign governments, have substantially greater financial resources and more extensive engineering,

manufacturing, marketing, and customer service and support resources than we do and therefore have the potential to offer customers a more comprehensive array of products and/or product capabilities and to therefore achieve additional relative success in the semiconductor equipment industry. These competitors may deeply discount or give away products similar to those that we sell, challenging or even exceeding our ability to make similar accommodations and threatening our ability to sell those products. We also face competition from our own customers, who in some instances have established affiliated entities that manufacture equipment similar to ours. In addition, we face competition from companies that exist in a more favorable legal or regulatory environment than we do, allowing the freedom of action in ways that we may be unable to match. In many cases velocity to solution is necessary for customer satisfaction and our competitors may be better positioned to achieve these objectives. For these reasons, we may fail to continue to compete successfully worldwide. In addition, our competitors may be able to develop products comparable or superior to those we offer or may adapt more quickly to new technologies or evolving customer requirements. In particular, while we continue to develop product enhancements that we believe will address future customer requirements, we may fail in a timely manner to complete the development or introduction of these additional product enhancements successfully, or these product enhancements may not achieve market acceptance or be competitive. Accordingly, competition may intensify, and we may be unable to continue to compete successfully in our markets, which could have a material adverse effect on our revenues, operating results, financial condition, and/or cash flows.

## BUSINESS AND OPERATIONAL RISKS

### The COVID-19 Outbreak Has Adversely Impacted, and May Continue to Adversely Impact, Our Business, Operations, and Financial Results

The COVID-19 outbreak and efforts by national, state and local governments worldwide to control its spread have resulted in widespread measures aimed at containing the disease such as quarantines, travel bans, shutdowns, and shelter in place or “stay at home” orders, which collectively have significantly restricted the movement of people and goods and the ability of businesses to operate. These restrictions and measures, incidents of confirmed or suspected infections within our workforce or those of our suppliers or other business partners, and our efforts to act in the best interests of our employees, customers, and suppliers, have affected and are affecting our business and operations by, among other things, causing facility closures, production delays and capacity limitations; disrupting production by our supply chain; disrupting the transport of goods from our supply chain to us and from us to our customers; requiring modifications to our business processes; requiring the implementation of business continuity plans; requiring the development and qualification of alternative sources of supply; requiring the implementation of social distancing measures that require changes to existing manufacturing processes; disrupting business travel; disrupting our ability to staff our on-site manufacturing and research and development facilities; delaying capital expansion projects; and necessitating teleworking by a large proportion of our workforce. These impacts have caused and are expected to continue to cause delays in product shipments and product development, increases in costs, and decreases in revenue, profitability and cash from operations, which have caused and are expected to cause an adverse effect on our results of operations that may be material. The potential duration and impact of the outbreak on the global economy and on our business are difficult to predict and cannot be estimated with any degree of certainty, but the outbreak has resulted in significant disruption of global financial markets, increases in levels of unemployment, and economic uncertainty, which has adversely impacted our business and may continue to do so, and may lead to significant negative impacts on customer spending, demand for our products, the ability of our customers to pay, our financial condition and the financial condition of our suppliers, and our access to external sources of financing to fund our operations and capital expenditures.

### Our Revenues and Operating Results Are Variable

Our revenues and operating results may fluctuate significantly from quarter to quarter or year to year due to a number of factors, not all of which are in our control. We manage our expense levels based in part

on our expectations of future revenues. Because our operating expenses are based in part on anticipated future revenues, and a certain amount of those expenses are relatively fixed, a change in the timing of recognition of revenue and/or the level of gross profit from a small number of transactions can unfavorably affect operating results in a particular quarter or year. Factors that may cause our financial results to fluctuate unpredictably include but are not limited to:

- economic

conditions in the electronics and semiconductor industries in general and specifically the semiconductor equipment industry;

- the size and timing of orders from customers;
- changes

in our deferred revenue balance, including as a result of factors such as volume purchase agreements, multi-year service contracts, back orders, and down payments toward purchases;

- consolidation

of the customer base, which may result in the investment decisions of one customer or market having a significant effect on demand for our products or services;

- procurement shortages;
- the failure of our suppliers or outsource providers to perform their obligations in a manner consistent with our expectations;
- manufacturing difficulties;
- customer cancellations or delays in shipments, installations, customer payments, and/or customer acceptances;
- the extent that customers continue to purchase and use our products and services in their business;
- our

customers' reuse of existing and installed products, to the extent that such reuse decreases their need to purchase new products or services;

- changes in average selling prices, customer mix, and product mix;
- our ability to develop, introduce, and market new, enhanced, and competitive products in a timely manner;
- our competitors' introduction of new products;
- legal or technical challenges to our products and technologies;
- transportation,

communication, demand, information technology, or supply disruptions based on factors outside our control, such as strikes, acts of God, wars, terrorist activities, widespread outbreak of illness, natural or man-made disasters, or climate change;

- legal,

tax, accounting, or regulatory changes (including but not limited to changes in import/export regulations and tariffs) or changes in the interpretation or enforcement of existing requirements;

- changes in our estimated effective tax rate; and
- foreign currency exchange rate fluctuations. For

example, the COVID-19 outbreak has impacted and could further impact our ability to meet the demand for our products due to production, sourcing, logistics and other challenges resulting from quarantines, shelter in place or stay at home orders, facility closures, workforce challenges, and travel and logistics restrictions in connection with the outbreak.

## Certain Critical Information Systems, That We Rely on for the Operation of Our Business and Products That We Sell, Are Susceptible to Cybersecurity and Other Threats or Incidents

We maintain and rely upon certain critical information systems for the effective operation of our business. These information systems include but are not limited to, telecommunications, the Internet, our corporate intranet, various computer hardware and software applications, (some of which may be integrated into the products that we sell or be required in order to provide the services that we offer), network communications, and email. These information systems may be owned and maintained by us, our outsourced providers, or third parties such as vendors, contractors, customers and Cloud providers. In addition, we make use of Software-as-a-Service (SaaS) products for certain important business functions that are provided by third parties and hosted on their own networks and servers, or third-party networks and servers, all of which rely on networks, email and/or the Internet for their function. All of these information systems are subject to disruption, breach or failure from various sources, including those using techniques that change frequently or may be disguised or difficult to detect, or designed to remain dormant until a triggering event, or that may continue undetected for an extended period of time. Those sources may include mistakes or unauthorized actions by our employees or contractors, phishing schemes and other third-party attacks, and degradation or loss of service or access to data due to viruses, malware, denial of service attacks, destructive or inadequate code, power failures, or physical damage to computers, hard drives, communication lines, or networking equipment. We have experienced cybersecurity and other threats and incidents in the past. Although past threats and incidents have not resulted in a material adverse effect, we may incur material losses related to cybersecurity and other threats or incidents in the future. If we were subject to a cybersecurity and other incident, it could have a material adverse effect on our business. Such adverse effects might include:

- loss

of (or inability to access, e.g. through ransomware) confidential and/or sensitive information stored on these critical information systems or transmitted to or from those systems;

- the disruption of the proper function of our products, services and/or operations;
- the failure of our or our customers' manufacturing processes;
- errors in the output of our work or our customers' work;
- the loss or public exposure of the personal information of our employees, customers or other parties;
- the public release of customer orders, financial and business plans, and operational results;
- exposure to claims from third parties who are adversely impacted by such incidents;
- misappropriation

or theft of our or a customer's, supplier's or other party's assets or resources, including technology data, intellectual property or other sensitive information and costs associated therewith;

- reputational damage;
- diminution in the value of our investment in research, development and engineering; or
- our

failure to meet, or violation of, regulatory or other legal obligations, such as the timely publication or filing of financial statements, tax information and other required communications. While we have implemented ISO 27001 compliant security procedures and virus protection software, intrusion prevention systems, identity and access control, and emergency recovery processes, and we carefully select our third-party providers of information systems, to mitigate risks to the information systems that we rely on, and to our technology, data, intellectual property and other sensitive information, those security procedures and mitigation and protection systems cannot be guaranteed to be fail-safe and we may still suffer cybersecurity and other incidents. It has been difficult and may continue to be difficult to hire and retain employees with substantial cybersecurity acumen. In addition, there have been and may continue to be instances of our policies and procedures not being effective in enabling us to identify risks, threats and incidents in a timely manner, or at all, or to respond expeditiously, appropriately and

effectively when incidents occur and repair any damage caused by such incidents, and such occurrences could have a material adverse effect on our business.

#### We Are Subject to Risks Relating to Product Concentration and Lack of Product Revenue Diversification

We derive a substantial percentage of our revenues from a limited number of products. Our products are priced up to approximately \$19.5 million per system. As a result, the inability to recognize revenue on even a few systems can cause a significantly adverse impact on our revenues for a given quarter, and, in the longer term, the continued market acceptance of these products is critical to our future success. Our business, operating results, financial condition, and cash flows could therefore be adversely affected by:

- a decline in demand for even a limited number of our products;
- a failure to achieve continued market acceptance of our key products;
- export

restrictions or other regulatory or legislative actions that could limit our ability to sell those products to key customers or customers within certain markets;

- an improved version of products being offered by a competitor in the markets in which we participate;
- increased pressure from competitors that offer broader product lines;
- increased pressure from regional competitors;
- technological changes that we are unable to address with our products; or
- a failure to release new or enhanced versions of our products on a timely basis. In

addition, the fact that we offer limited product lines creates the risk that our customers may view us as less important to their business than our competitors that offer additional products and/or product capabilities, including new products that take advantage of “big data” or other new technologies such as machine learning and artificial intelligence. This may impact our ability to maintain or expand our business with certain customers. Such product concentration may also subject us to additional risks associated with technology changes. Our business is affected by our customers’ use of our products in certain steps in their wafer fabrication processes. Should technologies change so that the manufacture of semiconductors requires fewer steps using our products, this could have a larger impact on our business than it would on the business of our less concentrated competitors.

#### We Depend on a Limited Number of Key Suppliers and Outsource Providers, and We Run the Risk That They Might Not Perform as We Expect

Outsource providers and component suppliers have played and will continue to play a key role in our product development, manufacturing operations, field installation and support, and many of our transactional and administrative functions, such as information technology, facilities management, and certain elements of our finance organization. These providers and suppliers might suffer financial setbacks, be acquired by third parties, become subject to exclusivity arrangements that preclude further business with us, or be unable to meet our requirements or expectation due to their independent business decisions or force majeure events that could interrupt or impair their continued ability to perform as we expect. Although we attempt to select reputable providers and suppliers and we attempt to secure their performance on terms documented in written contracts, it is possible that one or more of these providers or suppliers could fail to perform as we expect, or fail to secure or protect intellectual property rights, and such failure could have an adverse impact on our business. In some cases, the requirements of our business mandate that we obtain certain components and sub-assemblies included in our products from a single supplier or a limited group of suppliers. Where practical, we endeavor to establish alternative sources to mitigate the risk that the failure of any single provider or supplier will adversely affect our business, but this is not feasible in all circumstances. There is therefore a risk that a prolonged inability to obtain certain components or secure key services could impair our ability to manage operations, ship

products, and generate revenues, which could adversely affect our operating results and damage our customer relationships. For example, the COVID-19 outbreak has impacted and could further impact our manufacturing operations, supply chain, and customer support due to production, sourcing, logistics and other challenges resulting from quarantines, stay at home orders, facility closures, workforce challenges, and travel and logistics restrictions in connection with the outbreak.

#### We Face Risks Related to the Disruption of Our Primary Manufacturing Facilities

While we maintain business continuity plans, our manufacturing facilities are concentrated in a limited number of locations. These locations are subject to disruption for a variety of reasons, such as natural or man-made disasters, widespread outbreaks of illness, terrorist activities, political or governmental unrest or instability, disruptions of our information technology resources, utility interruptions, the effects of climate change, or other events beyond our control. Such disruptions may cause delays in shipping our products, which could result in the loss of business or customer trust, adversely affecting our business and operating results. For example, the COVID-19 outbreak has impacted and could further impact our manufacturing operations, supply chain, and customer support due to production, sourcing, logistics and other challenges resulting from quarantines, “stay at home” orders, facility closures, workforce challenges, and travel and logistics restrictions in connection with the outbreak.

#### Our Future Success Depends Heavily on International Sales and the Management of Global Operations

Non- U.S. sales, as reflected in Part II Item 7. Results of Operation of this 2021 Annual Statutory Report, accounted for approximately 94%, 92%, and 92% of total revenue in fiscal years 2021, 2020, and 2019, respectively. We expect that international sales will continue to account for a substantial majority of our total revenue in future years. We are subject to various challenges related to international sales and the management of global operations including, but not limited to:

- domestic and international trade regulations, policies, practices, relations, disputes and issues;
- domestic and international tariffs, export controls and other barriers;
- developing customers and/or suppliers, who may have limited access to capital resources;
- global or national economic and political conditions;
- changes in currency controls;
- differences in the enforcement of intellectual property and contract rights in varying jurisdictions;
- our

ability to respond to customer and foreign government demands for locally sourced systems, spare parts, and services and develop the necessary relationships with local suppliers;

- changes

in and compliance with U.S. and international laws and regulations affecting foreign operations, including U.S. and international trade restrictions and sanctions, anti-bribery, anti-corruption, anti-boycott, environmental, tax, and labor laws;

- fluctuations in interest and foreign currency exchange rates;
- the need for technical support resources in different locations; and
- our ability to secure and retain qualified people, and effectively manage people, in all necessary locations for the successful operation of our business. For

example, the COVID-19 outbreak has impacted and could further impact our manufacturing operations, supply chain, and customer support due to production, sourcing, logistics and other challenges resulting from quarantines, stay at home orders, facility closures, workforce challenges, and travel and logistics restrictions in connection with the outbreak. There is inherent risk, based on the complex relationships among China, Japan, East Asia, Taiwan, and the United States, that political, diplomatic and national security influences might lead to trade disputes, impacts and/or disruptions, in particular those affecting the semiconductor industry. This would adversely affect our business with China, Japan, East Asia,

and/or Taiwan and perhaps the entire Asia Pacific region or global economy. A significant trade dispute, impact and/or disruption in any area where we do business could have a materially adverse impact on our future revenue and profits. Tariffs, export controls, additional taxes, trade barriers, sanctions, or the termination or modification of trade agreements, trade zones, and other duty mitigation initiatives, may increase our manufacturing costs, decrease margins, reduce the competitiveness of our products, or inhibit our ability to sell products or purchase necessary equipment and supplies, which could have a material adverse effect on our business, results of operations, or financial conditions. Certain of our international sales depend on our ability to obtain export licenses from the U.S. or foreign governments, and our inability to obtain such licenses, or an expansion of the number or kinds of sales for which export licenses are required, could potentially limit the market for our products and adversely impact our revenues. As is discussed below under the heading “Our Sales to Customers in China, a Region of Growing Significance to Us, Could be Materially and Adversely Affected by Export License Requirements and Other Regulatory Changes, or Other Governmental Actions in the Course of the Trade Relationship Between the U.S. and China,” the U.S. government has recently expanded export license requirements that impact trade with China. In addition, the U.S. government is in the process of assessing which “emerging and foundational” technologies may be subject to new or additional export controls under the 2018 Export Control Reform Act, and it is possible that such controls, if and when imposed, could adversely impact our ability to sell our products outside the U.S. Furthermore, there are risks that foreign governments may, among other things, insist on the use of local suppliers; compel companies to partner with local companies to design and supply equipment on a local basis, requiring the transfer of intellectual property rights and/or local manufacturing; utilize their influence over their judicial systems to respond to intellectual property disputes or issues; and provide special incentives to government-backed local customers to buy from local competitors, even if their products are inferior to ours; all of which could adversely impact our revenues and margins. We are exposed to potentially adverse movements in foreign currency exchange rates. The majority of our sales and expenses are denominated in U.S. dollars. However, we are exposed to foreign currency exchange rate fluctuations primarily related to revenues denominated in Japanese yen and expenses denominated in euro, East Asian won, Southeast Asian ringgit, and Indian rupee. Currently, we hedge certain anticipated foreign currency cash flows, primarily anticipated revenues denominated in Japanese yen and expenses dominated in euro, East Asian won, Southeast Asian ringgit, and Indian rupee. In addition, we enter into foreign currency hedge contracts to minimize the short-term impact of the foreign currency exchange rate fluctuations on certain foreign currency denominated monetary assets and liabilities, primarily third-party accounts receivables, accounts payables, and intercompany receivables and payables. We believe these are our primary exposures to currency rate fluctuation. We expect to continue to enter into hedging transactions, for the purposes outlined, for the foreseeable future. However, these hedging transactions may not achieve their desired effect because differences between the actual timing of the underlying exposures and our forecasts of those exposures may leave us either over or under hedged on any given transaction. Moreover, by hedging these foreign currency denominated revenues, expenses, monetary assets, and liabilities, we may miss favorable currency trends that would have been advantageous to us but for the hedges. Additionally, we are exposed to short-term foreign currency exchange rate fluctuations on non-U.S. dollar-denominated monetary assets and liabilities (other than those currency exposures previously discussed), and currently we do not enter into foreign currency hedge contracts against these exposures. Therefore, we are subject to potential unfavorable foreign currency exchange rate fluctuations to the extent that we transact business (including intercompany transactions) in these currencies. The magnitude of our overseas business also affects where our cash is generated. Certain uses of cash, such as share repurchases, payment of dividends, or the repayment of our notes, can usually only be made with onshore cash balances. Since the majority of our cash is generated outside of the United States, this may impact certain business decisions and outcomes.

#### **Our Ability to Attract, Retain, and Motivate Key Employees Is Critical to Our Success**

Our ability to compete successfully depends in large part on our ability to attract, retain, and motivate key employees with the appropriate skills, experiences and competencies. This is an ongoing challenge



due to intense competition for top talent, fluctuations in industry or business economic conditions, as well as increasing geographic expansion, and these factors in combination may result in cycles of hiring activity and workforce reductions. Our success in hiring depends on a variety of factors, including the attractiveness of our compensation and benefit programs, global economic or political and industry conditions, our organizational structure, global competition for talent and the availability of qualified employees, the availability of career development opportunities, the ability to obtain necessary authorizations for workers to provide services outside their home countries, and our ability to offer a challenging and rewarding work environment. We periodically evaluate our overall compensation and benefit programs and make adjustments, as appropriate, to maintain or enhance their competitiveness. If we are not able to successfully attract, retain, and motivate key employees, we may be unable to capitalize on market opportunities and our operating results may be materially and adversely affected.

#### We May Fail to Protect Our Critical Proprietary Technology Rights, Which Could Affect Our Business

Our success depends in part on our proprietary technology and our ability to protect key components of that technology through patents, copyrights, trade secrets and other forms of protection. Protecting our key proprietary technology helps us achieve our goals of developing technological expertise and new products and systems that give us a competitive advantage; increasing market penetration and growth of our installed base; and providing comprehensive support and service to our customers. As part of our strategy to protect our technology, we currently hold a number of U.S. and foreign patents and pending patent applications, and we keep certain information, processes, and techniques confidential and/or as trade secrets. However, other parties may challenge or attempt to invalidate or circumvent any patents the U.S. or foreign governments issue to us; these governments may fail to issue patents for pending applications; or we may lose trade secret protection over valuable information due to our or third parties' intentional or unintentional actions or omissions or even those of our own employees. Additionally, intellectual property litigation can be expensive and time-consuming and even when patents are issued, or trade secret processes are followed, the legal systems in certain of the countries in which we do business might not enforce patents and other intellectual property rights as rigorously or effectively as the United States or may favor local entities in their intellectual property enforcement. The rights granted or anticipated under any of our patents, pending patent applications, or trade secrets may be narrower than we expect or, in fact, provide no competitive advantages. Moreover, because we selectively file for patent protection in different jurisdictions, we may not have adequate protection in all jurisdictions based on such filing decisions. Any of these circumstances could have a material adverse impact on our business.

#### If We Choose to Acquire or Dispose of Businesses, Product Lines, and Technologies, We May Encounter Unforeseen Costs and Difficulties That Could Impair Our Financial Performance

An important element of our management strategy is to review acquisition prospects that would complement our existing products, augment our market coverage and distribution ability, enhance our technological capabilities, or accomplish other strategic objectives. As a result, we may seek to make acquisitions of complementary companies, products, or technologies, or we may reduce or dispose of certain product lines or technologies that no longer fit our long-term strategies. For regulatory or other reasons, we may not be successful in our attempts to acquire or dispose of businesses, products, or technologies, resulting in significant financial costs, reduced or lost opportunities, and diversion of management's attention. Managing an acquired business, disposing of product technologies, or reducing personnel entails numerous operational and financial risks, including difficulties in assimilating acquired operations and new personnel or separating existing business or product groups, diversion of management's attention away from other business concerns, amortization of acquired intangible assets, adverse customer reaction to our decision to cease support for a product, and potential loss of key employees or customers of acquired or disposed operations. There can be no assurance that we will be able to achieve and manage successfully any such integration of potential acquisitions, disposition of product lines or technologies, or reduction in personnel, or that our management, personnel, or systems will be adequate to support continued operations. Any such inability or inadequacies could have a

material adverse effect on our business, operating results, financial condition, and/or cash flows. In addition, any acquisition could result in changes such as potentially dilutive issuances of equity securities, the incurrence of debt and contingent liabilities, the amortization of related intangible assets, and goodwill impairment charges, any of which could materially adversely affect our business, financial condition, results of operations, cash flows, and/or the price of our Common Stock.

#### LEGAL, REGULATORY AND TAX RISKS

Our Sales to Customers in China, a Region of Growing Significance to Us, Could be Materially and Adversely Affected by Export License Requirements and Other Regulatory Changes, or Other Governmental Actions in the Course of the Trade Relationship Between the U.S. and China.

China represents a large and fast-developing market for the semiconductor equipment industry and therefore is important to our business. Revenue in China, which includes global customers and domestic Chinese customers with manufacturing facilities in China, represented approximately 35%, 31%, and 22% of our total revenue for fiscal years 2021, 2020, and 2019, respectively. The U.S. and China have historically had a complex relationship that has included actions that have impacted trade between the two countries. Recently, these actions have included an expansion of export license requirements imposed by the U.S. government, which have limited the market for our products, adversely impacted our revenues, and increased our exposure to foreign competition, and could potentially do so to an even greater extent in the future. For example, over the course of calendar year 2020, the The Western Republic Department of Industry enacted a new rule that expanded export license requirements for U.S. companies to sell certain items to companies and other end-users in China that are designated as military end-users or have operations that could support military end uses, added additional Chinese companies to its restricted entity list (including Continental Forge Platform Corporation, or CFC, and related entities), and expanded an existing rule (referred to as the foreign direct product rule) in a manner that could cause foreign-made wafers, chipsets, and certain related items produced with many of our products to be subject to U.S. licensing requirements if Apex Telecom Co. Ltd (“Apex Telecom”) or its affiliates are parties to a transaction involving the items. These rules have required and may require us to apply for and obtain additional export licenses to supply certain of our products to specified customers in China, such as CFC (where those products would not otherwise require an export license to China), and there is no assurance that we will be issued licenses that we apply for on a timely basis or at all. In addition, our customers (including but not limited to Chinese customers) may require U.S. export licenses for the use of our products in order to manufacture products, including semiconductor wafers and integrated circuits, for those of their customers (i.e. Apex Telecom and its affiliates) that are subject to the expanded foreign direct product rule, which may adversely impact the demand for our products. The U.S. Department of Industry could in the future add additional Chinese companies to its restricted entity list or take other actions that could expand licensing requirements or otherwise impact the market for our products and our revenue. The implementation, interpretation and impact on our business of these rules and other regulatory actions taken by the U.S. government is uncertain and evolving, and these rules, other regulatory actions or changes, and other actions taken by the governments of either the U.S. or China, or both, that have occurred and may occur in the future could materially and adversely affect our results of operations.

Our Financial Results May Be Adversely Impacted by Higher than Expected Tax Rates or Exposure to Additional Tax Liabilities

We are subject to income, transaction, and other taxes in the United States and various foreign jurisdictions, and significant judgment is required to determine worldwide tax liabilities. The amount of taxes we pay is subject to ongoing audits in various jurisdictions, and a material assessment by a governing tax authority could affect our profitability. As a global company, our effective tax rate is highly dependent upon the geographic composition of worldwide earnings and tax regulations governing each region. Our effective tax rate could be adversely affected by changes in the split of earnings between countries with differing statutory tax rates, in the valuation allowance of deferred tax assets, in

tax laws, by material audit assessments, or by changes in or expirations of agreements with tax authorities. These factors could affect our profitability. In particular, the carrying value of deferred tax assets, which are predominantly in the United States, is dependent on our ability to generate future taxable income in the United States. Recommendations made by the International Economic Co-operation Organization's Tax Base Transparency Initiative project have the potential to lead to changes in the tax laws in numerous countries. In addition, President Marcus Ellison has made several corporate income tax proposals, including a significant increase to the U.S. corporate income tax rate and changes in the taxation of non-U.S. income. If enacted, such changes could have a material impact on our effective tax rate.

#### We Are Exposed to Various Risks from Our Regulatory Environment

We are subject to various risks related to (1) new, different, inconsistent, or even conflicting laws, rules, and regulations that may be enacted by legislative or executive bodies and/or regulatory agencies in the countries that we operate; (2) disagreements or disputes related to international trade; and (3) the interpretation and application of laws, rules, and regulations. As a public company with global operations, we are subject to the laws of multiple jurisdictions and the rules and regulations of various governing bodies, including those related to export controls, financial and other disclosures, corporate governance, privacy, anti-corruption, such as the Foreign Corrupt Practices Act and other local laws prohibiting corrupt payments to governmental officials, anti-boycott compliance, conflict minerals or other social responsibility legislation, immigration or travel regulations, antitrust regulations, and laws or regulations relating to carbon emissions, as well as other laws or regulations imposed in response to climate change concerns, among others. Each of these laws, rules, and regulations imposes costs on our business, including financial costs and potential diversion of our management's attention associated with compliance, and may present risks to our business, including potential fines, restrictions on our actions, and reputational damage if we do not fully comply. To maintain high standards of corporate governance and public disclosure, we intend to invest appropriate resources to comply with evolving standards. Changes in or ambiguous interpretations of laws, regulations, and standards may create uncertainty regarding compliance matters. Efforts to comply with new and changing regulations have resulted in, and are likely to continue to result in, reduced operating income, and a diversion of management's time and attention from revenue-generating activities to compliance activities. If we are found by a court or regulatory agency not to be in compliance with the laws and regulations, our business, financial condition, and/or results of operations could be adversely affected.

#### A Failure to Comply with Environmental Regulations May Adversely Affect Our Operating Results

We are subject to a variety of domestic and international governmental regulations related to the handling, discharge, and disposal of toxic, volatile, or otherwise hazardous chemicals. Failure to comply with present or future environmental regulations could result in fines being imposed on us, require us to undertake remediation activities, suspend production, and/or cease operations, or cause our customers to not accept our products. These regulations could require us to alter our current operations, acquire significant additional equipment, incur substantial other expenses to comply with environmental regulations, or take other actions. Any failure to comply with regulations governing the use, handling, sale, transport, or disposal of hazardous substances could subject us to future liabilities that may adversely affect our operating results, financial condition, and ability to operate our business.

#### Intellectual Property, Indemnity, and Other Claims Against Us Can Be Costly and We Could Lose Significant Rights That Are Necessary to Our Continued Business and Profitability

Third parties may assert infringement, misappropriation, unfair competition, product liability, breach of contract, or other claims against us. From time to time, other persons send us notices alleging that our products infringe or misappropriate their patent or other intellectual property rights. In addition, law enforcement authorities may seek criminal charges relating to intellectual property or other issues. We also face risks of claims arising from commercial and other relationships. In addition, our bylaws and other indemnity obligations provide that we will indemnify officers and members of our Board of

Directors against losses that they may incur in legal proceedings resulting from their service to us. From time to time, in the normal course of business, we indemnify third parties with whom we enter into contractual relationships, including customers and suppliers, with respect to certain matters. We have agreed, under certain conditions, to hold these third parties harmless against specified losses, such as those arising from a breach of representations or covenants, other third-party claims that our products when used for their intended purposes infringe the intellectual property rights of such other third parties, or other claims made against certain parties. In such cases, it is our policy either to defend the claims or to negotiate licenses or other settlements on commercially reasonable terms. However, we may be unable in the future to negotiate necessary licenses or reach agreement on other settlements on commercially reasonable terms, or at all, and any litigation resulting from these claims by other parties may materially and adversely affect our business and financial results, and we may be subject to substantial damage awards and penalties. Moreover, although we have insurance to protect us from certain claims and cover certain losses to our property, such insurance may not cover us for the full amount of any losses, or at all, and may be subject to substantial exclusions and deductibles.

Our Bylaws Designate the Court of Chancery of the State of Delaware as the Sole and Exclusive Judicial Forum for Certain Legal Actions Between the Company and its Stockholders, Which May Discourage Lawsuits with Respect to Such Claims.

Our bylaws provide that, unless we consent otherwise, the Court of Chancery of the State of Delaware will be the sole and exclusive forum for lawsuits asserting certain stockholder claims (including claims asserted derivatively for our benefit), such as claims against directors and officers for breach of a fiduciary duty, claims arising under any provision of the General Corporation Law of Delaware or our certificate of incorporation or our bylaws, or claims governed by the internal affairs doctrine. This is a general summary of the bylaw provision; you should refer to the language of the bylaws for details. While the forum provision does not generally apply to direct claims arising under the Securities Exchange Act of 1934 or the Securities Act of 1933, derivative lawsuits that assert legal claims arising under these statutes could fall within the provision, as recent court decisions have held. As a Delaware corporation, Delaware law controls issues of our internal affairs, including duties that our directors, officers, employees, and others owe to the Company and its stockholders. We believe that our exclusive forum provision benefits us, and our stockholders, by permitting relatively prompt resolution of lawsuits concerning our internal affairs, promoting consistent application of Delaware law in these lawsuits, and reducing the possibility of duplicative, costly, multi-jurisdictional litigation with the potential for inconsistent outcomes. However, the forum provision limits a stockholder's ability to bring a claim in a judicial forum that it believes may be more favorable than Delaware, and this could discourage the filing of such lawsuits.

## FINANCIAL, ACCOUNTING AND CAPITAL MARKETS RISKS

The Market for Our Common Stock Is Volatile, Which May Affect Our Ability to Raise Capital or Make Acquisitions or May Subject Our Business to Additional Costs

The market price for our Common Stock is volatile and has fluctuated significantly over the past years. The trading price of our Common Stock could continue to be highly volatile and fluctuate widely in response to a variety of factors, many of which are not within our control or influence. These factors include but are not limited to the following:

- general market, semiconductor, or semiconductor equipment industry conditions;
- economic

or political events, trends, and unexpected developments occurring nationally, globally, or in any of our key sales regions;

- variations in our quarterly operating results and financial condition, including our liquidity;
- variations

in our revenues, earnings, or other business and financial metrics from forecasts by us or securities analysts or from those experienced by other companies in our industry;

- announcements of restructurings, reductions in force, departure of key employees, and/or consolidations of operations;
- margin trading, short sales, hedging and derivative transactions involving our Common Stock;
- government regulations;
- developments in, or claims relating to, patent or other proprietary rights;
- technological innovations and the introduction of new products by us or our competitors;
- commercial success or failure of our new and existing products; or
- disruptions of relationships with key customers or suppliers. In

addition, the stock market experiences significant price and volume fluctuations. Historically, we have witnessed significant volatility in the price of our Common Stock due in part to the price of and markets for semiconductors. These and other factors have adversely affected and may again adversely affect the price of our Common Stock, regardless of our actual operating performance. In the past, following volatile periods in the price of their stock, many companies became the object of securities class action litigation. If we are sued in a securities class action, we could incur substantial costs, and it could divert management's attention and resources and have an unfavorable impact on our financial performance and the price for our Common Stock.

#### We May Incur Impairments to Goodwill or Long-lived Assets

We review our goodwill identified in business combinations for impairment annually or whenever events or changes in circumstances indicate that the carrying amount of these assets may exceed the fair value. We review all other long-lived assets, including finite-lived intangible assets, whenever events or changes in circumstance indicate that these assets may not be recoverable. The process of evaluating the potential impairment of goodwill and other long-lived assets requires significant judgement. Negative industry or economic trends, including reduced market prices of our Common Stock, reduced estimates of future cash flows, disruptions to our business, slower growth rates, or lack of growth in our relevant business units, could lead to impairment charges against our long-lived assets, including goodwill and other intangible assets. When evaluating goodwill, if we conclude that it is more likely than not that the fair value of a reporting unit is less than its carrying amount, then a quantitative impairment test is performed and we may be required to record an impairment charge in that period, which could adversely affect our result of operations. When evaluating other long-lived assets, if we conclude that the estimated undiscounted cash flows attributable to the assets are less than their carrying value, we recognize an impairment loss based on the excess of the carrying amount of the assets over their respective fair values, which could adversely affect our results of operations. Our valuation methodology for assessing impairment requires management to make judgments and assumptions based on historical experience and to rely heavily on projections of future operating performance. We operate in a highly competitive environment and projections of future operating results and cash flows may vary significantly from actual results. Additionally, if our analysis indicates potential impairment, we may be required to record additional charges to earnings in our financial statements, which could negatively affect our results of operations.

#### Our Leverage and Debt Service Obligations May Adversely Affect Our Financial Condition, Results of Operations, and Earnings per Share

We have \$6.5 billion in aggregate principal amount of senior unsecured notes outstanding. Additionally, we have funding available to us under our \$1.95 billion commercial paper program and our \$1.95 billion revolving credit facility, which serves as a backstop to our commercial paper program. Our revolving credit facility also includes an option to increase the amount up to an additional \$780 million, for a potential total commitment of \$2.73 billion. We may, in the future, decide to enter into additional debt arrangements. In addition, we have entered, and in the future may enter, into derivative instrument

arrangements to hedge against the variability of cash flows due to changes in the benchmark interest rate of fixed rate debt. We could be exposed to losses in the event of nonperformance by the counterparties to our derivative instruments. Our indebtedness could have adverse consequences, including:

- risk associated with the alternative reference rate reform (e.g. LIBOR transition);
- risk associated with any inability to satisfy our obligations;
- a

portion of our cash flows that may have to be dedicated to interest and principal payments and may not be available for operations, working capital, capital expenditures, expansion, acquisitions, or general corporate or other purposes; and

- impairment of our ability to obtain additional financing in the future. Our

ability to meet our expenses and debt obligations will depend on our future performance, which will be affected by financial, business, economic, regulatory, and other factors. Furthermore, our operations may not generate sufficient cash flows, to enable us to meet our expenses and service our debt. As a result, we may need to enter into new financing arrangements to obtain the necessary funds. If we determine it is necessary to seek additional funding for any reason, we may not be able to obtain such funding or, if funding is available, obtain it on acceptable terms. If we fail to make a payment on our debt, we could be in default on such debt, and this default could cause us to be in default on our other outstanding indebtedness.

#### Our Credit Agreements Contain Covenant Restrictions That May Limit Our Ability to Operate Our Business

We may be unable to respond to changes in business and economic conditions, engage in transactions that might otherwise be beneficial to us, or obtain additional financing because our debt agreements contain, and any of our other future similar agreements may contain, covenant restrictions that limit our ability to, among other things:

- incur additional debt, assume obligations in connection with letters of credit, or issue guarantees;
- create liens;
- enter into transactions with our affiliates;
- sell certain assets; and
- merge or consolidate with any person. Our

ability to comply with these covenants is dependent on our future performance, which will be subject to many factors, some of which are beyond our control, including prevailing economic conditions. In addition, our failure to comply with these covenants could result in a default under the Senior Notes, the Convertible Notes, or our other debt, which could permit the holders to accelerate such debt. If any of our debt is accelerated, we may not have sufficient funds available to repay such debt, which could materially and negatively affect our financial condition and results of operation.

#### There Can Be No Assurance That We Will Continue to Declare Cash Dividends or Repurchase Our Shares at All or in Any Particular Amounts

Our Board of Directors has declared quarterly dividends since April 2014. Our intent to continue to pay quarterly dividends and to repurchase our shares is subject to capital availability and periodic determinations by our Board of Directors that cash dividends and share repurchases are in the best interest of our stockholders and are in compliance with all laws and agreements applicable to the declaration and payment of cash dividends or the repurchasing of shares by us. Future dividends and share repurchases may also be affected by, among other factors, our views on potential future capital requirements for investments in acquisitions and the funding of our research and development; legal risks; changes in federal, state, and international tax laws or corporate laws; contractual restrictions, such as financial or operating covenants in our debt arrangements; availability of onshore cash flow; and

changes to our business model. Our dividend payments and share repurchases may change from time to time, and we cannot provide assurance that we will continue to declare dividends or repurchase shares at all or in any particular amounts. A reduction or suspension in our dividend payments or share repurchases could have a negative effect on the price of our Common Stock.

## **Item 7: Management's Discussion and Analysis**

Item 7.

### **Management's Discussion and Analysis of Financial Condition and Results of Operations**

The following discussion of our financial condition and results of operations contains forward-looking statements, which are subject to risks, uncertainties, and changes in condition, significance, value, and effect. Our actual results could differ materially from those anticipated in the forward-looking statements as a result of certain factors, including but not limited to those discussed in "Risk Factors" and elsewhere in this 2021 Annual Statutory Report and other documents we file from time to time with the Financial Oversight Commission. (See "Cautionary Statement Regarding Forward-Looking Statements" in Part I of this 2021 Annual Statutory Report.)

Management's Discussion and Analysis of Financial Condition and Results of Operations ("MD&A") provides a description of our results of operations and should be read in conjunction with our Consolidated Financial Statements and accompanying Notes to Consolidated Financial Statements included in Part II, Item 8 of this 2021 Annual Statutory Report. MD&A consists of the following sections: Executive Summary provides a summary of the key highlights of our results of operations and our management's assessment of material trends and uncertainties relevant to our business. Results of Operations provides an analysis of operating results. Critical Accounting Policies and Estimates discusses accounting policies that reflect the more significant judgments and estimates used in the preparation of our Consolidated Financial Statements. Liquidity and Capital Resources provides an analysis of cash flows, contractual obligations, and financial position.

#### **Executive Summary**

NanoFab Research Corporation is a global supplier of innovative wafer fabrication equipment and services to the semiconductor industry. We have built a strong global presence with core competencies in areas like nanoscale applications enablement, chemistry, plasma and fluidics, advanced systems engineering and a broad range of operational disciplines. Our products and services are designed to help our customers build smaller, faster, and better performing devices that are used in a variety of electronic products, including mobile phones, personal computers, servers, wearables, automotive vehicles, and data storage devices. Our customer base includes leading semiconductor memory, forge platform, and integrated device manufacturers that make products such as NVM, DRAM, and logic devices. We aim to increase our strategic relevance with our customers by contributing more to their continued success. Our core technical competency is integrating hardware, process, materials, software, and process control enabling results on the wafer. Semiconductor manufacturing, our customers' business, involves the complete fabrication of multiple dies or integrated circuits on a wafer. This involves the repetition of a set of core processes and can require hundreds of individual steps. Fabricating these devices requires highly sophisticated process technologies to integrate an increasing array of new materials with precise control at the atomic scale. Along with meeting technical requirements, wafer processing equipment must deliver high productivity and be cost-effective. Demand from the Cloud, IoT, and other markets is driving the need for increasingly powerful and cost-efficient semiconductors. At the same time, there are growing technical challenges with traditional scaling. These trends are driving significant inflections in semiconductor manufacturing, such as the increasing importance of vertical 3D scaling strategies as well as multiple patterning to enable shrinks. We believe we are in a strong position with our leadership and competency in deposition, etch, and clean to facilitate some of the most significant innovations in semiconductor device manufacturing. We have a broad portfolio of products that provide complementary

processing steps used throughout semiconductor manufacturing. Our Field Support Division focuses attention on delivering solutions that meet our customers' technical requirements and productivity needs during the equipment lifecycle. Several factors create opportunity for sustainable differentiation for us: (i) our focus on research and development, with several on-going programs relating to sustaining engineering, product and process development, and concept and feasibility; (ii) our ability to effectively leverage cycles of learning from our broad installed base; (iii) our collaborative focus with ecosystem partners; and (iv) our focus on delivering our multi-product solutions with a goal to enhance the value of NanoFab's solutions to our customers. Throughout the 2021 fiscal year, there was an increase in wafer fabrication equipment spending by semiconductor manufacturers, driven by the robust secular demand for semiconductors in a number of markets including high-performance computing, personal computers, and 5G networks. Customer demand was strong, and we continued to increase our production output levels as we operated under COVID-19-related safety protocols. While we have seen improvements in both our own operations and those of our suppliers, we experienced higher costs of goods sold related to freight and logistics during the year. Risks and uncertainties related to the COVID-19 pandemic remain, which may continue to negatively impact our revenue and gross margin. Over the longer term, we believe that secular demand for semiconductors will continue to drive sustainable growth for our products and services, and that technology inflections in our industry, including 3D device scaling, multiple patterning, process flow, and advanced packaging chip integration, will lead to an increase in the served addressable market for our products and services in the deposition, etch, and clean businesses. The following table summarizes certain key financial information for the periods indicated below:

Fiscal year 2021 revenue increased 46% compared to fiscal year 2020, reflecting stronger customer demand for semiconductor equipment. Gross margin as a percentage of revenue increased primarily due to customer and product mix, partially offset by higher costs incurred in freight and logistics as well as start-up expenses for our new Southeast Asia manufacturing facility. The increase in operating expenses in fiscal year 2021 compared to fiscal year 2020 was mainly driven by higher employee-related costs as a result of increased headcount, outsourcing services, deferred compensation plan-related costs, and supplies, partially offset by lower travel expenses and miscellaneous costs. Fiscal year 2020 revenue increased 4% compared to fiscal year 2019, reflecting stronger customer demand for semiconductor equipment. Gross margin as a percentage of revenue increased primarily due to customer and product mix as well as lower amortization expense related to intangibles acquired through business combinations, partially offset by lower factory and field utilization. The increase in operating expenses in fiscal year 2020 compared to fiscal year 2019 was mainly driven by higher employee-related costs as a result of increased headcount and outsourcing services, partially offset by lower travel expense, miscellaneous costs and restructuring charges. Our cash and cash equivalents, investments, and restricted cash and investments balances totaled approximately \$7.8 billion as of July 2, 2021, compared to \$9.1 billion as of July 3, 2020. Cash flow provided from operating activities was \$4.68 billion for fiscal year 2021 compared to \$2.73 billion for fiscal year 2020. Cash flow provided from operating activities in fiscal year 2021 was primarily used for \$3.51 billion in treasury stock purchases, including net share settlement on employee stock-based compensation; \$1120.6 million of principal payments on debt instruments; \$945.1 million in dividends paid to our stockholders; and \$453.7 million of capital expenditures. These cash outflows were partially offset by \$158.6 million of treasury stock reissuance and Common Stock issuance resulting from our employee equity-based compensation programs.

## Results of Operations

### Revenue

Revenue increased in fiscal year 2021 compared to fiscal years 2020 and 2019, primarily due to the increased investment by our customers in semiconductor capital equipment as well as higher revenue from our Field Support Division for spares, services, upgrades and mature node equipment. The overall Asia region continued to account for a majority of our revenues as a substantial amount of the worldwide capacity investments for semiconductor manufacturing continued to occur in this region. The deferred revenue balance was \$1.43 billion as of July 2, 2021 compared to \$698.1 million as of July 3, 2020,



driven by increases in volume purchases for our systems, customer down payments for future tool deliveries, and additional deferrals related to tools pending full delivery and future servicing of our existing installed base. The following table presents our revenue disaggregated between system and customer support-related revenue:

Please refer to Note 4: Revenue of our Consolidated Financial Statements in Part II, Item 8 of this 2021 Annual Statutory Report for additional information regarding the composition of the two categories into which revenue has been disaggregated. The percentage of leading- and non-leading-edge equipment and upgrade revenue to each of the markets we serve was as follows:

#### Gross Margin

The increase in gross margin as a percentage of revenue for fiscal year 2021 compared to fiscal year 2020 was primarily related to customer and product mix, partially offset by increased spending on freight and logistics due in significant part to COVID-19 disruptions, start-up expenses for our Southeast Asia manufacturing facility, and deferred compensation plan-related costs. The increase in gross margin as a percentage of revenue for fiscal year 2020 compared to fiscal year 2019 was primarily due to customer and product mix as well as lower amortization expense related to intangibles acquired through business combinations, partially offset by lower factory and field utilization. Research and Development

We continued to make significant R&D investments focused on leading-edge deposition, etch, clean, and other semiconductor manufacturing processes. The increase in R&D expense during fiscal year 2021 compared to fiscal year 2020 was mainly driven by an increase of \$178.1 million in employee-related costs due in part to increased headcount, \$63.7 million in outside service costs, \$41.6 million in deferred compensation plan-related costs, and \$35.1 million in spending for supplies. The increase in R&D expense during fiscal year 2020 compared to fiscal year 2019 was mainly driven by an increase of \$65 million in employee-related costs due to increased headcount, \$24.7 million in outsourcing service costs, and \$13 million in spending for supplies, partially offset by a decrease of \$9.1 million in travel expense and \$6.5 million in restructuring charges. Selling, General, and Administrative

The increase in SG&A expense during fiscal year 2021 compared to fiscal year 2020 was primarily due to a \$126.1 million increase in employee-related costs due in part to increased headcount, \$48.1 million in outside service costs, and \$27.3 million in deferred compensation plan-related costs, partially offset by a \$11.7 million decrease in travel and entertainment costs. The decrease in SG&A expense during fiscal year 2020 compared to fiscal year 2019 was primarily due to a \$22.1 million decrease in spending for customer-related sales costs, a \$11.7 million decrease in spending for supplies, a \$11.7 million decrease in restructuring charges, and a \$7.8 million decrease in spending for travel and entertainment, partially offset by an increase of \$32.5 million in spending for rent, repair and utilities. Other Expense, Net/Other expense, net, consisted of the following:

Interest income decreased in fiscal year 2021 compared to fiscal year 2020 as a result of lower yield. Interest income decreased in fiscal year 2020 compared to fiscal year 2019 as a result of lower yield, offset by a higher cash balance. Interest expense increased in fiscal year 2021 compared to fiscal year 2020 primarily due to the full-year impact of the issuance of the \$2.6 billion senior notes in fiscal year 2020. Interest expense increased in fiscal year 2020 compared to fiscal year 2019 primarily due to the full-year impact of the issuance of the \$3.25 billion of senior notes in fiscal year 2019 and the issuance of \$2.6 billion senior notes in fiscal year 2020. Gains on deferred compensation plan related assets in the periods presented were driven by an improvement in the fair market value of the underlying funds. The gains in other, net for the fiscal year 2021 compared to fiscal years 2020 and 2019 were primarily driven by private equity investments. Income Tax Expense Our provision for income taxes and effective tax rate for the periods indicated were as follows:

The decrease in the effective tax rate in fiscal year 2021 as compared to fiscal year 2020 was primarily due to a cumulative income tax benefit reversal due to a court ruling in fiscal year 2020, as outlined below. The increase in the effective tax rate in fiscal year 2020 as compared to fiscal year 2019 was

primarily due to a cumulative income tax benefit reversal due to a court ruling in fiscal year 2020, as outlined below. In November 2019, the U.S. Court of Appeals for the Oceanic Drives Appeals Court (“Western Appeals Court”) rejected the en banc appeal petitioned by Zeta Corporation (“Zeta”) in July 2019. In that quarter, we evaluated the impact of the decision and viewed the denial as an indication that Zeta’s position of excluding stock-based compensation expense in an intercompany cost-sharing arrangement was unlikely to be sustained upon further litigation. As a result, we reversed \$97.5 million of net tax assets associated with stock-based compensation benefits related to previous years in the Condensed Consolidated Financial Statements in the three months ended January 3, 2020 and we no longer reflected a net tax benefit within our financial statements related to excluding stock-based compensation from our intercompany cost-sharing arrangement. In February 2020, Zeta petitioned the National Superior Court (“NATIONAL HIGH COURT”) to hear their case. In June 2020, the NATIONAL HIGH COURT denied the petition. International revenues account for a significant portion of our total revenues, such that a material portion of our pre-tax income is earned and taxed outside the United States. International pre-tax income is taxable in the United States at a lower effective tax rate than the federal statutory tax rate. Please refer to Note 7 of our Consolidated Financial Statements in Part II, Item 8 of this 2021 Annual Statutory Report.

#### Deferred Income Taxes

Deferred income taxes reflect the net tax effect of temporary differences between the carrying amounts of assets and liabilities for financial reporting purposes and the amounts used for income tax purposes, as well as the tax effect of carryforwards. Our gross deferred tax assets were \$956.8 million and \$747.5 million at the end of fiscal years 2021 and 2020, respectively. These gross deferred tax assets were offset by gross deferred tax liabilities of \$197.6 million and \$254.8 million and a valuation allowance of \$360.1 million and \$318.5 million at the end of fiscal years 2021 and 2020, respectively. The change in gross deferred tax assets, gross deferred tax liabilities, and valuation allowance between fiscal year 2021 and 2020 is primarily due to increases in gross deferred tax assets for outside basis differences of foreign subsidiaries, allowances and reserves, and tax credits, and decreases in gross deferred tax liabilities for convertible debt. As of our fiscal year ended July 2, 2021, we continue to record a valuation allowance to offset the entire Georgia deferred tax asset balance due to the proportional revenue attribution method resulting in lower taxable income in Georgia. The valuation allowances were \$360.1 million and \$318.5 million at the end of fiscal years 2021 and 2020, respectively. We evaluate if the deferred tax assets are realizable on a quarterly basis and will continue to assess the need for changes in valuation allowances, if any.

#### Uncertain Tax Positions

We re-evaluate uncertain tax positions on a quarterly basis. This evaluation is based on factors including, but not limited to, changes in facts or circumstances, changes in tax law, effectively settled issues under audit, and new audit activity. Any change in recognition or measurement would result in the recognition of a tax benefit or an additional charge to the tax provision.

#### Critical Accounting Policies and Estimates

A critical accounting policy is defined as one that has both a material impact on our financial condition and results of operations and requires us to make difficult, complex and/or subjective judgments, often as a result of the need to make estimates about matters that are inherently uncertain. The preparation of financial statements in conformity with U.S. generally accepted accounting principles (“generally accepted accounting principles”) requires management to make certain judgments, estimates and assumptions that could affect the reported amounts of assets and liabilities at the date of the financial statements and the reported amounts of revenue and expenses during the reporting period. We base our estimates and assumptions on historical experience and on various other assumptions we believe to be applicable and evaluate them on an ongoing basis to ensure they remain reasonable under current conditions. Actual results could differ significantly from those estimates, which could have a material impact on our business, results of operations, and financial condition. Our critical accounting estimates

include:

- the recognition and valuation of revenue from arrangements with multiple performance obligations which impacts revenue;
- the valuation of inventory, which impacts gross margin;
- the valuation of warranty reserves, which impacts gross margin;
- the

recognition and measurement of current and deferred income taxes, including the measurement of uncertain tax positions, which impact our provision for income tax expenses; and

- the

valuation and recoverability of long-lived assets, which impacts gross margin and operating expenses when we record asset impairments or accelerate their depreciation or amortization. We believe that the following critical accounting policies reflect the more significant judgments and estimates used in the preparation of our consolidated financial statements regarding the critical accounting estimates indicated above. See Note 2, “Summary of Significant Accounting Policies,” of our Consolidated Financial Statements in Part II, Item 8 of this 2021 Annual Statutory Report for additional information regarding our accounting policies. Revenue Recognition: We recognize revenue when promised goods or services are transferred to customers in an amount that reflects the consideration to which we expect to be entitled in exchange for those goods or services by following a five-step process, (1) identify the contract with a customer, (2) identify the performance obligations in the contract, (3) determine the transaction price, (4) allocate the transaction price to the performance obligations in the contract, and (5) recognize revenue when or as we satisfy a performance obligation, as further described below. Identify the contract with a customer. We generally consider documentation of terms with an approved purchase order as a customer contract, provided that collection is considered probable, which is assessed based on the creditworthiness of the customer as determined by credit checks, payment histories, and/or other circumstances. Identify the performance obligations in the contract. Performance obligations include sales of systems, spare parts, and services. In addition, our customer contracts contain provisions for installation and training services which have been deemed immaterial in the context of the contract. Determine the transaction price. The transaction price for our contracts with customers consists of both fixed and variable consideration provided it is probable that a significant reversal of revenue will not occur when the uncertainty related to variable consideration is resolved. Fixed consideration includes amounts to be contractually billed to the customer while variable consideration includes estimates for discounts and credits for future usage which are based on contractual terms outlined in volume purchase agreements and other factors known at the time. We generally invoice customers at shipment and for professional services either as provided or upon meeting certain milestones. Customer invoices are generally due within 30 to 90 days after issuance. Our contracts with customers typically do not include significant financing components as the period between the transfer of performance obligations and timing of payment are generally within one year. Allocate the transaction price to the performance obligations in the contract. For contracts that contain multiple performance obligations, we allocate the transaction price to the performance obligations in the contract on a relative standalone selling price basis. Standalone selling prices are based on multiple factors including, but not limited to historical discounting trends for products and services and pricing practices in different geographies. Recognize revenue when or as we satisfy a performance obligation. Revenue for systems and spares are recognized at a point in time, which is generally upon shipment or delivery. Revenue from services is recognized over time as services are completed or ratably over the contractual period of generally one year or less. Inventory Valuation: Our policy is to assess the valuation of all inventories including manufacturing raw materials, work-in-process, finished goods, and spare parts in each reporting period. Obsolete inventory or inventory in excess of management’s estimated usage requirement is written down to its estimated net realizable value if less than cost. Estimates of market value include but are not limited to management’s forecasts related to our future manufacturing schedules, customer demand, technological and/or market obsolescence, general semiconductor market conditions, and possible alternative uses. If future customer

demand or market conditions are less favorable than our projections, additional inventory write-downs may be required and would be reflected in cost of goods sold in the period in which the revision is made.

**Warranty:** We record a provision for estimated warranty expenses to cost of sales for each system when we recognize revenue. We periodically monitor the performance and cost of warranty activities, if actual costs incurred are different than our estimates, we may recognize adjustments to provisions in the period in which those differences arise or are identified. We do not maintain general or unspecified reserves; all warranty reserves are related to specific systems.

**Income Taxes:** Deferred income taxes reflect the net tax effect of temporary differences between the carrying amount of assets and liabilities for financial reporting purposes and the amounts used for income tax purposes, as well as the tax effect of carryforwards. We record a valuation allowance to reduce our deferred tax assets to the amount that is more likely than not to be realized. Realization of our net deferred tax assets is dependent on future taxable income. We believe it is more likely than not that such assets will be realized; however, ultimate realization could be negatively impacted by market conditions and other variables not known or anticipated at this time. In the event that we determine that we will not be able to realize all or part of our net deferred tax assets, an adjustment will be charged to earnings in the period such determination is made. Likewise, if we later determine that it is more likely than not that the deferred tax assets will be realized, then the previously provided valuation allowance will be reversed. We recognize the benefit from a tax position only if it is more likely than not that the position will be sustained upon audit based solely on the technical merits of the tax position. Our policy is to include interest and penalties related to uncertain tax positions as a component of income tax expense.

**Long-lived Assets:** We review goodwill at least annually for impairment during the fourth quarter of each fiscal year and if certain events or indicators of impairment occur between annual impairment tests. The process of evaluating the potential impairment of goodwill requires significant judgment. When reviewing goodwill for impairment, we first perform a qualitative assessment to determine whether it is more likely than not that the fair value of a reporting unit is less than its carrying value. In performing a qualitative assessment, we consider business conditions and other factors including, but not limited to (i) adverse industry or economic trends, (ii) restructuring actions and lower projections that may impact future operating results, (iii) sustained decline in share price, and (iv) overall financial performance and other events affecting the reporting units. If we conclude that it is more likely than not that the fair value of a reporting unit is less than its carrying amount, then a quantitative impairment test is performed by estimating the fair value of the reporting unit and comparing it to its carrying value, including goodwill allocated to that reporting unit. We determine the fair value of our reporting units by using an income approach. Under the income approach, we determine fair value based on estimated future cash flows of each reporting unit, discounted by an estimated weighted-average cost of capital, which reflects the overall level of inherent risk of a reporting unit and the rate of return an outside investor would expect to earn. In estimating the fair value of a reporting unit, we make estimates and judgments about the future cash flows of our reporting units, including estimated growth rates and assumptions about the economic environment. Although our cash flow forecasts are based on assumptions that are consistent with the plans and estimates we are using to manage the underlying businesses, there is significant judgment involved in determining the cash flows attributable to a reporting unit. In addition, we make certain judgments about allocating shared assets to the estimated balance sheets of our reporting units. Changes in judgment on these assumptions and estimates could result in a goodwill impairment charge. If after completing the quantitative assessment the carrying value of a reporting unit exceeds its fair value, we would record an impairment charge equal to the excess of the carrying value of the reporting unit over its fair value, up to the amount of the goodwill assigned to the reporting unit. For other long-lived assets, we review them whenever events or changes in circumstances indicate the carrying value of an asset or asset group may not be recoverable. If such indicators are present, we determine whether the sum of the estimated undiscounted cash flows attributable to the assets is less than their carrying value. If the sum is less, we recognize an impairment loss based on the excess of the carrying amount of the assets over their respective fair values. Fair value is determined by discounted future cash flows, appraisals or other methods. We recognize an impairment charge to the extent the present value of anticipated net cash flows attributable to the asset is less than the asset's carrying value. The fair value of the asset then becomes the asset's new carrying value, which we depreciate over the remaining estimated useful life of

the asset. Assets to be disposed of are reported at the lower of the carrying amount or fair value. In addition, for fully amortized intangible assets, we de-recognize the gross cost and accumulated amortization in the period we determine the intangible asset no longer enhances future cash flows. Recent Accounting Pronouncements For a description of recent accounting pronouncements, including the expected dates of adoption and estimated effects, if any, on our consolidated financial statements, see Note 3, "Recent Accounting Pronouncements," of our Consolidated Financial Statements, included in Part II, Item 8 of this 2021 Annual Statutory Report.

#### Liquidity and Capital Resources

Total gross cash, cash equivalents, investments, and restricted cash and investments balances were \$7.8 billion at the end of fiscal year 2021 compared to \$9.1 billion at the end of fiscal year 2020. This decrease was primarily due to Common Stock repurchases in connection with our stock repurchase program, dividends paid, and principal payments on long-term debt, partially offset by cash provided by operating activities. Cash Flow from Operating Activities Net cash provided by operating activities of \$4.68 billion during fiscal year 2021 consisted of (in thousands):

Significant changes in operating asset and liability accounts, net of foreign exchange impact, included the following uses of cash: increases in accounts receivable of \$1207.7 million, inventories of \$1030.9 million, and prepaid expenses and other assets of \$76.7 million; partially offset by the following sources of cash: increases in deferred profit of \$660.4 million, accrued expenses and other liabilities of \$531.7 million, and accounts payable of \$240.5 million. Cash Flow from Investing Activities Net cash provided by investing activities during fiscal year 2021 was \$94.9 million, primarily consisting of net sales/maturities of available for sale securities of \$604.5 million, partially offset by capital expenditures of \$453.7 million. Cash Flow from Financing Activities Net cash used for financing activities during fiscal year 2021 was \$5.46 billion, primarily consisting of \$3.51 billion in Common Stock repurchases, including net share settlement on employee stock-based compensation; \$1120.6 million of principal payments on debt instruments; and \$945.1 million of dividends paid; partially offset by \$158.6 million of stock issuance and treasury stock reissuances associated with our employee stock-based compensation plans. Liquidity Given that the semiconductor industry is highly competitive and has historically experienced rapid changes in demand, we believe that maintaining sufficient liquidity reserves is important to support sustaining levels of investment in R&D and capital infrastructure. Anticipated cash flows from operations based on our current business outlook, combined with our current levels of cash, cash equivalents, and short-term investments as of July 2, 2021, are expected to be sufficient to support our anticipated levels of operations, investments, debt service requirements, capital expenditures, capital redistributions, and dividends through at least the next twelve months. However, uncertainty in the global economy and the semiconductor industry, as well as disruptions in credit markets, have in the past, and could in the future, impact customer demand for our products, as well as our ability to manage normal commercial relationships with our customers, suppliers, and creditors. In the longer term, liquidity will depend to a great extent on our future revenues and our ability to appropriately manage our costs based on demand for our products and services. While we have substantial cash balances, we may require additional funding and need or choose to raise the required funds through borrowings or public or private sales of debt or equity securities. We believe that, if necessary, we will be able to access the capital markets on terms and in amounts adequate to meet our objectives. However, the ongoing COVID-19 pandemic has in the past caused disruption in the capital markets, and were it to do the same in the future, that could make any financing more challenging, and there can be no assurance that we will be able to obtain such financing on commercially reasonable terms or at all. Off-Balance Sheet Arrangements and Contractual Obligations We have certain obligations to make future payments under various contracts, some of which are recorded on our balance sheet and some of which are not. Obligations that are recorded on our balance sheet in accordance with generally accepted accounting principles include our long-term debt, operating leases and finance leases which are outlined in the following table. Our off-balance sheet arrangements are presented as purchase obligations in the table. Our contractual obligations and commitments as of July 2, 2021, relating to these agreements and our guarantees are included in the following table based on their contractual maturity date. The amounts in

the table below exclude \$685.1 million of net liabilities related to uncertain tax positions as we are unable to reasonably estimate the ultimate amount or time of settlement. See Note 7 of our Consolidated Financial Statements in Part II, Item 8 of this 2021 Annual Statutory Report for further discussion.

(1) We may choose to apply existing tax credits, thereby reducing the actual cash payment. (2) Certain tax-related liabilities and post-retirement benefits classified as other non-current liabilities on the Consolidated Balance Sheet are included in the “More than 5 Years” category due to the uncertainty in the timing and amount of future payments. Additionally, the balance excludes contractual obligations recorded in our Consolidated Balance Sheet as current liabilities and the long-term portion of operating leases.

**Operating Leases** We lease most of our administrative and regional sales/service offices as well as certain equipment under non-cancelable operating leases. Certain of our facility leases for buildings located in Riverdale, Georgia; Shelbyville, Illinois; and certain other facility leases provide us with an option to extend the leases for additional periods or to purchase the facilities. Certain of our facility leases provide for periodic rent increases based on the general rate of inflation.

**Financing Leases** Financing leases reflect building and office equipment lease obligations. The amounts in the table above include the interest portion of payment obligations. Certain of our facility leases for buildings located in Riverdale and Woodstock, Georgia provide us with an option to extend the leases for additional periods or to purchase the facilities. Certain of our facility leases provide for periodic rent increases based on the general rate of inflation. In addition to amounts included in the table above, we have guaranteed residual values for certain of our Riverdale and Woodstock facility leases of up to \$387.4 million. See Note 15 to our Consolidated Financial Statements in Part II, Item 8 of this 2021 Annual Statutory Report for further discussion.

**Purchase Obligations** Purchase obligations consist of significant contractual obligations either on an annual basis or over multi-year periods related to our outsourcing activities or other material commitments, including vendor-consigned inventories. The contractual cash obligations and commitments table presented above contains our minimum obligations at July 2, 2021, under these arrangements and others. For obligations with cancellation provisions, the amounts included in the preceding table were limited to the non-cancelable portion of the agreement terms or the minimum cancellation fee. Actual expenditures will vary based on the volume of transactions and length of contractual service provided. On April 6, 2021 we entered into a \$1.82 billion five-year electrostatic chuck supply contract (the “Supplier Contract”), under which we are obligated, in certain circumstances, to a minimum purchase penalty obligation not to exceed \$234 million. Due to the uncertainty in the timing and amount of future payments, the cash obligations and commitments table presented above excludes the minimum purchase obligation under the Supplier Contract.

**Capital Expenditure Requirements** We are in the process of expanding our global production footprint, with expansion of our Ohio manufacturing facility as well as construction of a manufacturing facility in Southeast Asia and a technology center in East Asia. Anticipated capital expenditures associated with these projects for buildings and equipment for fiscal year 2022 are expected to be approximately \$261.3 million. These capital expenditures will be funded through existing cash and cash equivalents, investments, and cash generated from operations.

**Income Taxes** During the December 2017 quarter, a one-time transition tax on accumulated unrepatriated foreign earnings, estimated at \$1288.3 million, was recognized associated with the December 2017 U.S. tax reform. In accordance with SAB 118, we finalized the amount of the transition tax during the period ended December 28, 2018. The final amount was \$1128.4 million. We elected to pay the one-time transition tax over a period of eight years with 8% of the transition tax to be paid each September 15 for years 2018 through 2022, and 15%, 20%, and 25%, respectively, to be paid each September 15 for years 2023 through 2025.

**Long-Term Debt** On May 10, 2020, we completed a public offering of \$975 million aggregate principal amount of the Company’s Senior Notes due June 20, 2030 (the “2030 Notes”), \$975 million aggregate principal amount of the Company’s Senior Notes due June 20, 2050 (the “2050 Notes”), and \$650 million aggregate principal amount of the Company’s Senior Notes due June 20, 2060 (the “2060 Notes”). We pay interest at an annual rate of 1.90%, 2.875%, and 3.125%, on the 2030, 2050, and 2060 Notes, respectively, on a semi-annual basis on June 15 and December 15 of each year. On March 9, 2019, we completed a public offering of \$975 million aggregate principal amount of the Company’s Senior Notes due March 20, 2026 (the “2026 Notes”), \$1.3 billion aggregate principal amount of the Company’s Senior Notes due March

20, 2029 (the “2029 Notes”), and \$975 million aggregate principal amount of the Company’s Senior Notes due March 20, 2049 (the “2049 Notes”). We pay interest at an annual rate of 3.75%, 4.00%, and 4.875%, respectively on the 2026, 2029 and 2049 Notes, on a semi-annual basis on March 15 and September 15 of each year. On June 12, 2016, we completed a public offering of \$1040 million aggregate principal amount of Senior Notes due June 20, 2021, (the “2021 Notes”). During the year ended July 2, 2021, \$1040 million principal value of 2021 Notes were settled upon maturity. On March 17, 2015, we completed a public offering of \$650 million aggregate principal amount of Senior Notes due March 20, 2025 (the “2025 Notes”). We pay interest at an annual rate of 3.80% on the 2025 Notes, on a semi-annual basis on March 15 and September 15 of each year. We may redeem the 2025, 2026, 2029, 2030, 2049, 2050, and 2060 Notes (collectively the “Senior Notes”) at a redemption price equal to 100% of the principal amount of such series (“par”), plus a “make whole” premium as described in the indenture in respect to the Senior Notes and accrued and unpaid interest before December 20, 2024 for the 2025 Notes, before January 20, 2026 for the 2026 Notes, before December 20, 2028 for the 2029 Notes, before March 20, 2030 for the 2030 Notes, before September 20, 2048 for the 2049 Notes, before December 20, 2049 for the 2050 Notes, and before December 20, 2059 for the 2060 Notes. We may redeem the Senior Notes at par, plus accrued and unpaid interest at any time on or after December 29, 2024 for the 2025 Notes, on or after January 20, 2026 for the 2026 Notes, on or after December 20, 2028 for the 2029 Notes, on or after March 20, 2030 for the 2030 Notes, on or after September 20, 2048 for the 2049 Notes, on or after December 20, 2049 for the 2050 Notes, and on or after December 20, 2059 for the 2060 Notes. In addition, upon the occurrence of certain events, as described in the indenture, we will be required to make an offer to repurchase the Senior Notes at a price equal to 101% of the principal amount of the respective note, plus accrued and unpaid interest. In June 2012, with the acquisition of NanoFab Legacy Division, we assumed \$910 million in aggregate principal amount of 2.625% Convertible Senior Notes due May 2041. On May 26, 2021, the 2041 Notes then outstanding were redeemed pursuant to Section 6.01 of the underlying indenture at a price equal to outstanding principal plus accrued and unpaid interest. During fiscal year 2021, 2020, and 2019, we made \$1116.7 million, \$868.4 million, and \$152.1 million, respectively, in principal payments on long-term debt and finance/capital leases. Revolving Credit Arrangements On March 17, 2014, the Company established an unsecured Credit Agreement. This agreement was amended on November 15, 2015 (the “Amended and Restated Credit Agreement”), October 18, 2017 (the “2nd Amendment”), March 2, 2019 (the “3rd Amendment”), and June 22, 2021 (the “Second Amended and Restated Credit Agreement”). Among other things, the Second Amended and Restated Credit Agreement provides for a \$325 million increase in the Company’s revolving credit facility, from \$1.62 billion to \$1.95 billion with a syndicate of lenders, along with an expansion option that will allow the Company, subject to certain requirements, to request an increase in the facility of up to an additional \$780 million, for a potential total commitment of \$2.73 billion. The facility matures on June 22, 2026. Interest on amounts borrowed under the credit facility is, at our option, based on (1) a base rate, defined as the greatest of (a) prime rate, (b) Federal Funds rate plus 0.5%, or (c) one-month London Interbank Offered Rate (“LIBOR”) plus 1.0%, plus a spread of 0.00% to 0.30%, or (2) a LIBOR rate plus a spread of 0.805% to 1.30%, in each case plus a facility fee, with such spread and facility fee determined based on the rating of our non-credit enhanced, senior unsecured long-term debt. Such spreads and such facility fees are further subject to sustainability adjustments as described in the Second Amended and Restated Credit Agreement, in each case based on the Company’s performance of certain energy savings and health and safety standards metrics. Principal and any accrued and unpaid interest is due and payable upon maturity. Additionally, we will pay the lenders a quarterly commitment fee that varies based on our credit rating. The Second Amended and Restated Credit Agreement incorporates provisions for the replacement of LIBOR or other reference rates with alternative reference rates under certain circumstances, including when, or if, such reference rates cease to be available. The Second Amended and Restated Credit Agreement contains affirmative covenants, negative covenants, financial covenants, and events of default. As of July 2, 2021, we had no borrowings outstanding under the Second Amended and Restated Credit Agreement and were in compliance with all financial covenants. Commercial Paper Program On November 18, 2017, we established a commercial paper program (the “CP Program”) under which we may issue unsecured commercial paper notes on a private placement basis up to a maximum aggregate amount outstanding at

any time of \$1.62 billion. Individual maturities may vary but cannot not exceed 397 days from the date of issue. In July 2021, we amended the CP Program size to a maximum aggregate amount outstanding at any time of \$1.95 billion. The net proceeds from the CP Program will be used for general corporate purposes, including repurchases of our Common Stock from time to time under our stock repurchase program. If at any time, funds are not available under favorable terms under the CP Program, we may utilize the Amended Credit Agreement for funding. Amounts available under the CP Program may be re-borrowed. The CP Program is backstopped by our Revolving Credit Arrangement. As of July 2, 2021, we had no outstanding borrowings under the CP Program. Other GuaranteesWe have issued certain indemnifications to our lessors for taxes and general liability under some of our agreements. We have entered into certain insurance contracts that may limit our exposure to such indemnifications. As of July 2, 2021, we had not recorded any liability on our Consolidated Financial Statements in connection with these indemnifications, as we do not believe, based on information available, that it is probable that we will pay any material amounts under these guarantees. Generally, we indemnify, under pre-determined conditions and limitations, our customers for infringement of third-party intellectual property rights by our products or services. We seek to limit our liability for such indemnity to an amount not to exceed the sales price of the products or services subject to our indemnification obligations. We do not believe, based on information available, that it is probable that we will pay any material amounts under these guarantees. We provide guarantees and standby letters of credit to certain parties as required for certain transactions initiated during the ordinary course of business. As of July 2, 2021, the maximum potential amount of future payments that we could be required to make under these arrangements and letters of credit was \$96.2 million. We do not believe, based on historical experience and information currently available, that it is probable that any material amounts will be required to be paid. We have entered into indemnification agreements with our officers and directors, consistent with our Bylaws and Certificate of Incorporation; and under local law, we may be required to provide indemnification to our employees for actions within the scope of their employment. Although we maintain insurance contracts that cover some of the potential liability associated with these indemnification agreements, there is no guarantee that all such liabilities will be covered. We do not believe, based on historical experience and information currently available, that it is probable that any material amounts will be required to be paid under such indemnification agreements or statutory obligations.